



Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

MVC 2027 Organizers

Binh-Minh Nguyen¹, Facundo Aguilera²

¹The University of Tokyo, ²National University of Río Cuarto

IEEE VTS Motor Vehicles

Ricardo de Castro³

³University of California, Merced

子曰：「學而時習之，不亦說乎？有朋自遠方來，不亦樂乎？」

Confucius said: “Is it not a pleasure to study and constantly apply the lessons that you have learned? Is it not a joy to have friends visit from afar?”



With researchers from Italy



With researcher from Lithuania



Student internship from France

We would like to share with you our dream of the future e-mobility, and a vehicle simulator that we have been using *to teach* students and *to collaborate* with friends and colleagues around the world.

Agenda

☒ Introduction

☐ Motivation

☐ Problem setting of MVC2027

☐ MATLAB/Simulink files

☐ Demonstration of MESO strategies

☐ GitHub repository

☐ Registration and Submission

IEEE VPPC 2026

Lyon, 5-9 October 2026

Introduction to IEEE VTS Motor Vehicle Challenge

Ricardo de Castro, VTS VP Motor Vehicles

MV Challenge: What is it?

Competition to simulate development of energy management algorithms for Electric Vehicles

Mission Profile

Competitors

- Team one or more students/researchers

Competitor's Algorithm

Electric Vehicle **Simulation Model**
(*Matlab/Simulink*)

Key Performance Metrics (KPIs)

Final Score

Tasks:

- Sizing of energy storage/powertrain
- Energy management of Energy storage
- Torque allocation (front/rear) and vectoring (left/right)
- Planning of EV charging
- ...

Examples:

- Electric road vehicle
- Electric train
- Electric boat
- Electric truck
- ...

KPIs:

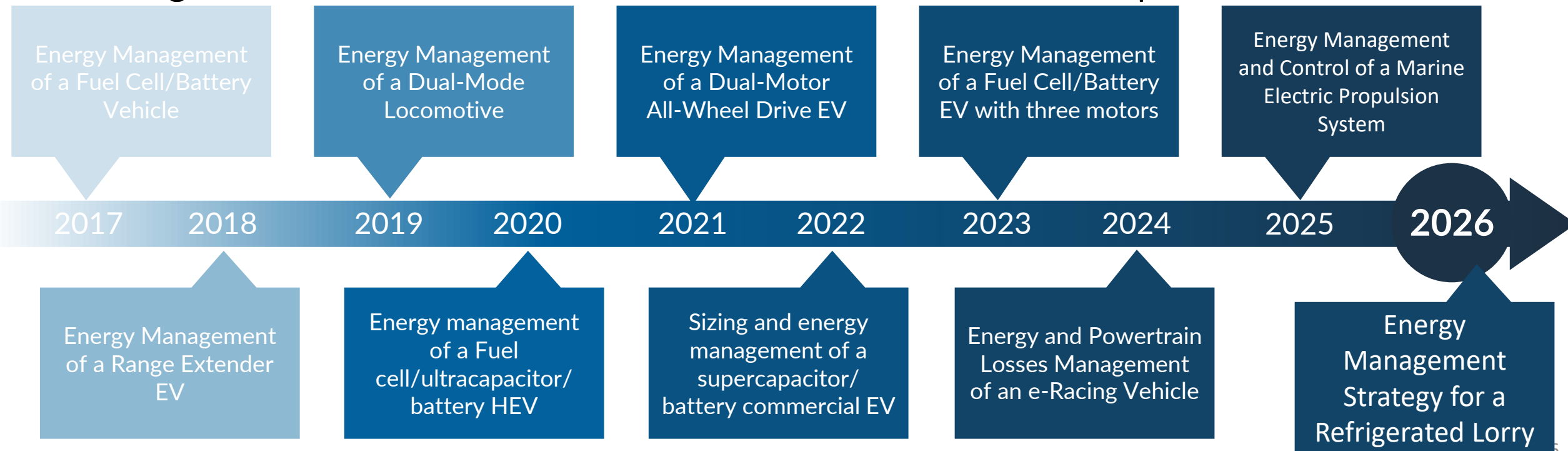
- Economic cost (\$)
- Energy consumption
- Degradation
- ...



Provided by VTS Organizers

MV Challenge: Previous editions

- First edition in 2016
- This year we are celebrating the **10th** edition
- Organization included institutions from North America to Europe and Asia

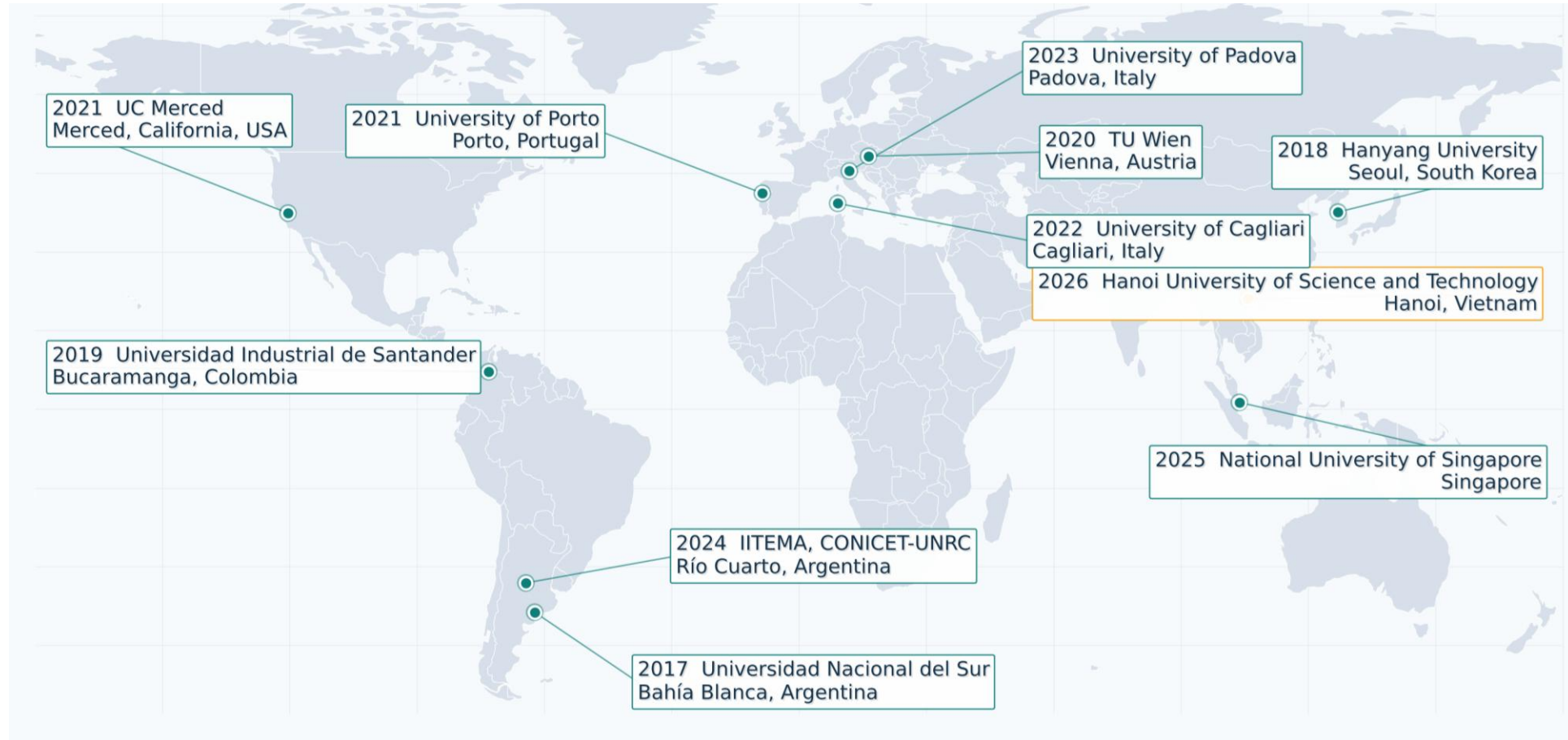


MV Challenge: Prizes

- Registration to the competition is **totally free**, but to eventually receive the award VTS membership is mandatory
- All participants will receive
 - Certificate of participation to the competition
 - Invitation to submit paper for the next VPPC edition
- Prizes for the **top two** teams
 - First prize: Up to a limit of 3500 US\$
 - Second prize: Up to a limit of 1500 US\$

Restriction: award covers
travel expenses to VPPC

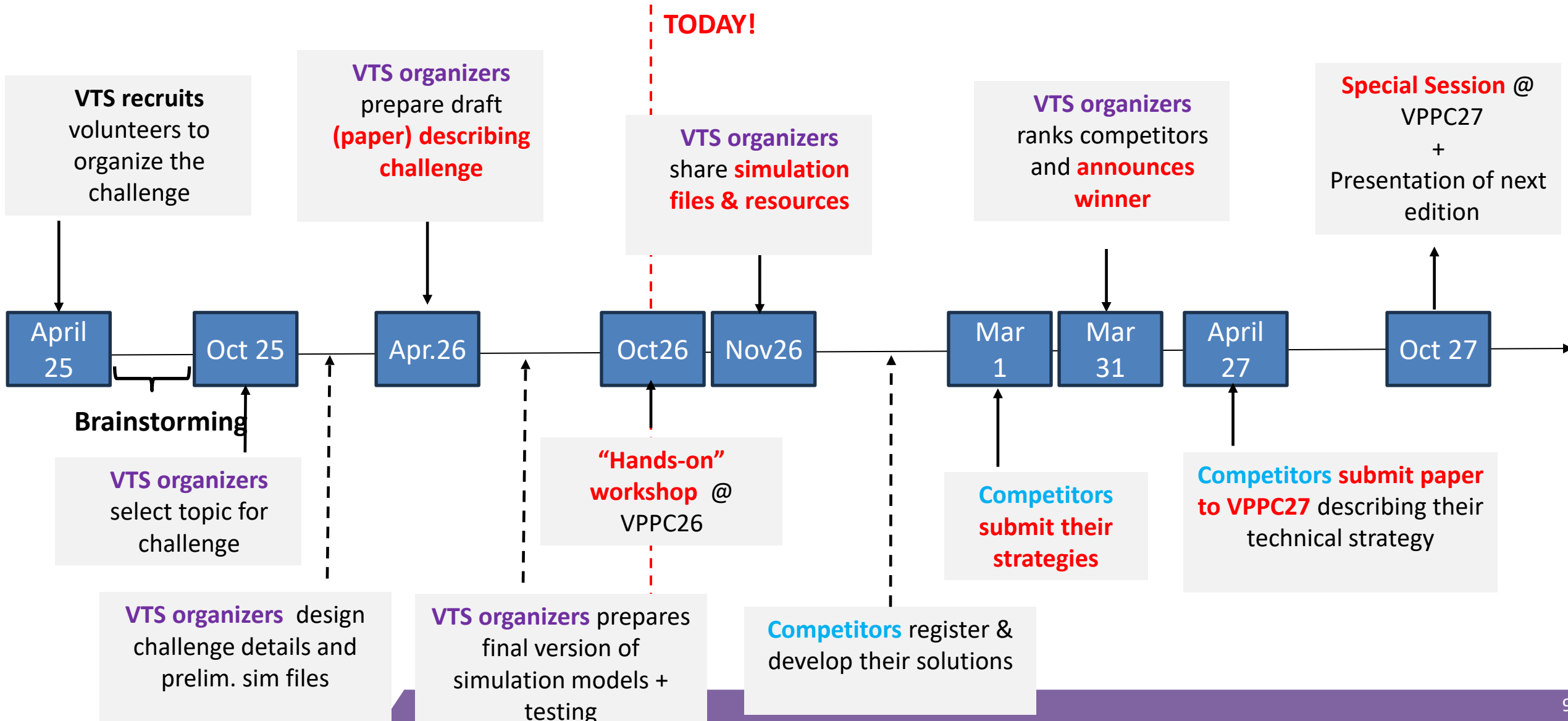
MV Challenge: Past Winners



**> 600 Researchers participated
in the competition**

> \$40k in prizes since 2016

Timeline and Milestones for 2027 Edition



MVC: Github Platform

Code Repository

Version Management

Q&A with competitors

The screenshot displays the GitHub interface for the **IEEE-MVC-2023** repository. The repository is public and has 5 watchers. The main branch is selected, showing 1 branch and 0 tags. The file structure on the left includes:

- Functions: Final Release of MVC 2023, p
- MVC2023_SIM_FMU: MVC Beta 3 version, based o
- MVCExportedFMUs: Final Release of MVC 2023, p
- Reporting: Final Release of MVC 2023, p
- Tracks: Updated preview information
- media: Udated ranking document
- .gitattributes: MVC Beta 3 version, based o
- .gitignore: Setup repository

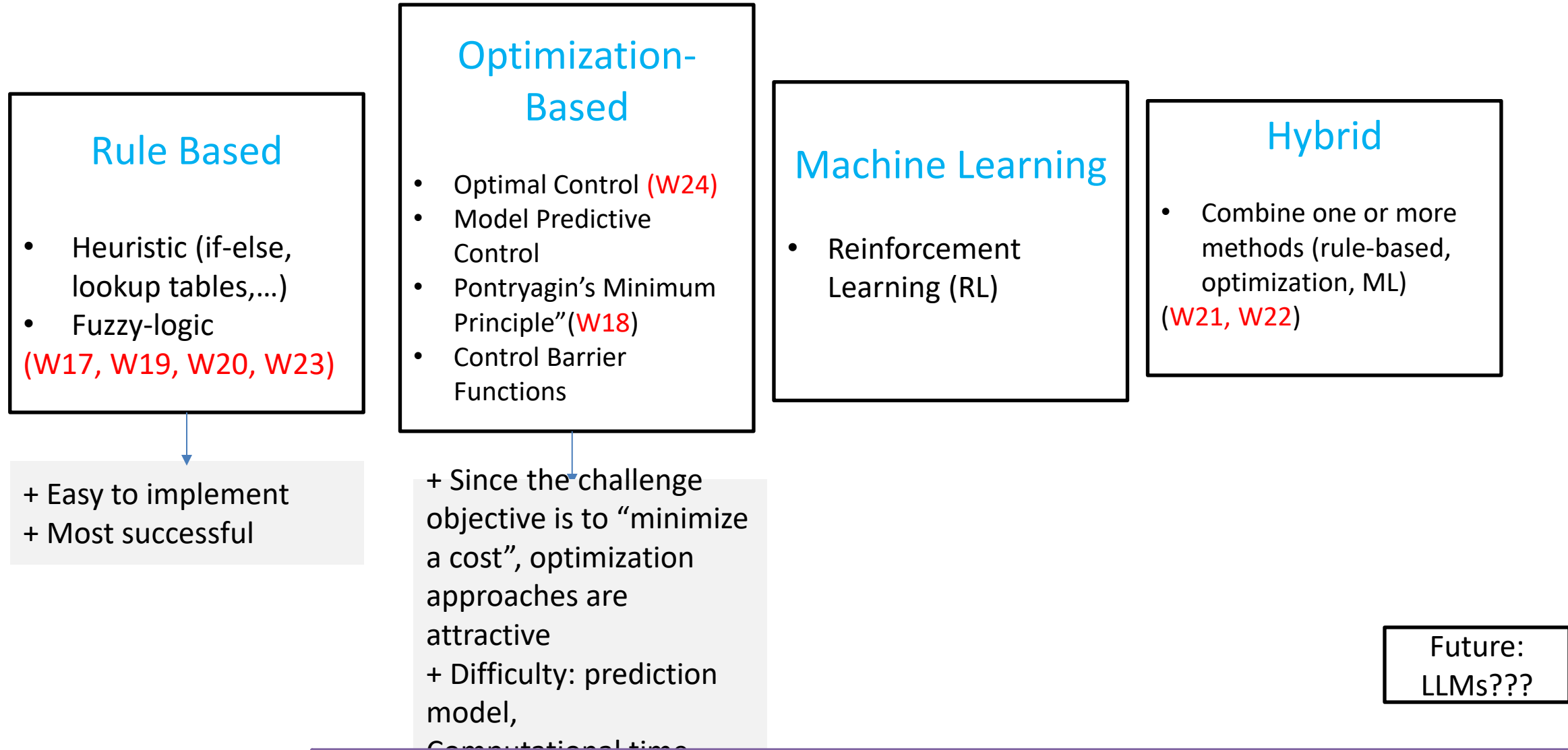
The commit history shows three recent commits:

- Updated BLScores.mat to match new preview horizon data** (JBRDLR committed on Feb 28, 2023)
- Updated preview information of tracks to have the same horizon length for all tracks; th** (JBRDLR committed on Feb 28, 2023)
- Update SUBGUIDE.md** (JBRDLR authored on Feb 21, 2023)

The Q&A section on the right lists several issues:

- Side slip angle issue** (woojmn asked on Feb 2, 2023 in Q&A - Unanswered)
- Side slip angle issue** (JBRDLR asked on Feb 2, 2023 in Q&A - Unanswered)
- [ERROR] Matlab/Simulink error** (woojmn asked on Dec 27, 2022 in Q&A - Answered)
- Git .mat and .slx files not downloaded in MATLAB** (grisamgit asked on Feb 1, 2023 in Q&A - Answered)
- Derating and temperature violation cost functions** (mariagrazia-tristano asked on Feb 1, 2023 in Q&A - Unanswered)

MVC: Control Strategies Used in Past Editions



MVC @ VPPC 2026

THURSDAY, 8 October; 11:00 am - 12:40 pm; Ampère A

WS2: Workshop on Motor Vehicle Challenge 2027 - Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

Binh Minh Nguyen, Facundo Aguilera, Ricardo de Castro | The Univ. of Tokyo, National Univ. of Río Cuarto, Univ. of California | Japan, Argentina, United States

2:00 pm - 2:20 pm: VTS Motor Vehicle Challenge 2027: Management of Energy Efficiency and Safe Operation for Multi-actuated Ground Vehicles on DWPT Road

Binh Minh Nguyen, Facundo Aguilera, Alexander González, Antonio Tota, Yuki Hosomi, Hiroshi Fujimoto | The Univ. of Tokyo, National Univ. of Río Cuarto, Politecnico di Torino | Japan, Argentina, Italy

2:20 pm - 2:40 pm: Design of Powertrain and Energy Management Strategy for a Refrigerated Lorry - IEEE VTS Challenge 2026

Tien Dat Chu, Hoang-Phuc Dam, Van Nghia Le, Trong-Thuan Tran, Ngoc Tu Vi, Anh Duc Nguyen, Huu Duc Vu | Hanoi Univ. of Science and Technology | Vietnam

2:40 pm - 3:00 pm: Integrated Energy Management for Electrified Refrigerated Heavy-Duty Vehicles With Torque Split and Charging Control

Marzio Barresi, Lorenza Cafarelli, Alessandra Casiraghi, Samuele Grillo | Politecnico di Milano | Italy

THURSDAY, 8 October; 2:00 - 4:00 pm; Ampère C

Session Chairs: Ludovico Ortombina (Univ. of Padova, Italy) & Fabio Tinazzi (Univ. of Padova, Italy)

SS12: IEEE VTS Motor Vehicle Challenge 2026 – Design of Powertrain and Energy Management Strategy for a Refrigerated Lorry

3:00 pm - 3:20 pm: Combined Genetic-Algorithm Charging Strategy and Fuzzy Energy Management for the IEEE VTS Motor Vehicle Challenge 2026

Alexandre Ravey, Chafic El Hajjar, Meiling Yue, Xicheng Xie, Ruomei Sun, Meng Lin | Univ. of Technology of Belfort Montbéliard, Beijing Jiaotong Univ. | France, China

3:20 pm - 3:40 pm: Energy Management Strategy for Electric Refrigerated Lorries via State Estimation with Speed Optimization and Charging Strategy

Changqi Yang, Yijia Peng, Sheng Zhao, Xiaodong Wu | Shanghai Jiao Tong Univ. | China

3:40 pm - 4:00 pm: A Hierarchical Collaborative Energy Management Strategy for Dual-Motor Refrigerated Trucks with Predictive Charging Planning and Loss-Optimal Torque Distribution

Cihang Wang, Tailiang Huang, Xianxu Bai, Chengkang Shao, Jie Li | Hefei Univ. of Technology, Anhui Polytechnic Univ. | China

MVC: Final Remarks

Questions/Comments?

- We are always looking for **volunteers** to
 - **Test** preliminary versions of MVC simulation files
 - **Organize** future MVC (ideas for new topics?)
- Future ideas/directions
 - Increase integration of MVC challenges into **course/capstone projects**
 - We may need to release simulation files in summer to better align with fall/spring semesters
 - Recruit **industry members** to co-organize future MVC challenges
 - **Summer school** on Motor Vehicle Challenge
 - First edition @ University of Rabat, Morocco in 2026.
 - *Extended Syllabus*: modeling, numerical simulation, algorithms (rule-based, optimization, machine learning,...), hands-on implementation activities with past MVC simulation,
 - **Anyone with interest in hosting a 2027 edition?**
 - New problems/**topics relevant for future of electric and sustainable transportation** (avoid repetition of problem already addressed in previous MVC editions.

Feel free to reach out to us if you would like to be involved (rpintodecastro@ucmerced.edu)

Questions?

Agenda

☐ Introduction

☒ **Motivation**

☐ Problem setting of MVC2027

☐ MATLAB/Simulink files

☐ Demonstration of MESO strategies

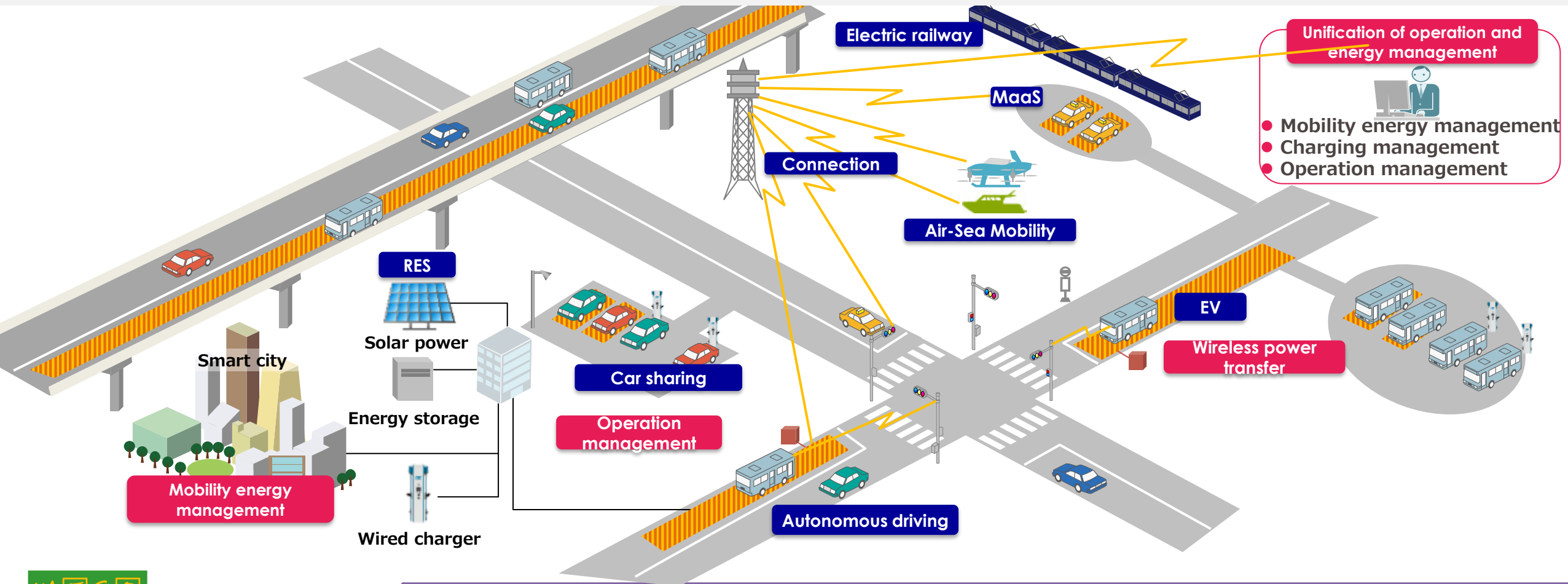
☐ GitHub repository

☐ Registration and Submission

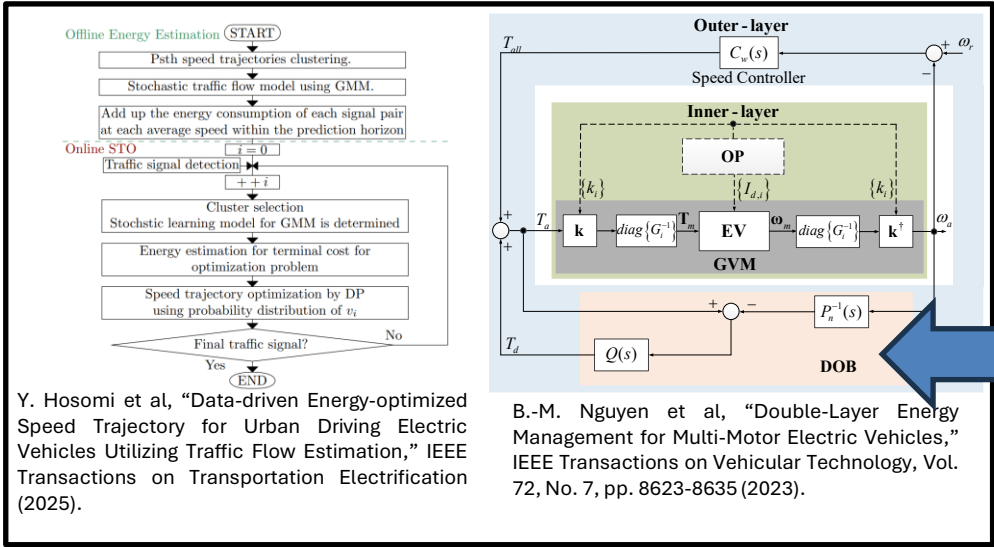
The future of e-Mobility

The transportation sector is undergoing a fundamental transformation driven by the urgent need to reduce carbon emissions and develop sustainable urban mobility systems.

- ❖ Emerging technologies such as *autonomous driving and motion control*, *optimal energy management and operation management*, and *dynamic wireless power transfer* are shaping the future of mobility.
- ❖ The development and integration of these systems demand not only technological innovation but also a *transdisciplinary approach* integrating expertise in electrical, automotive, control, mechanical, material engineering, etc.

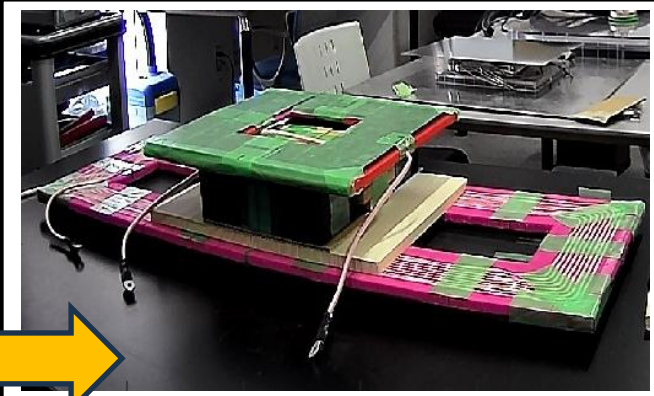
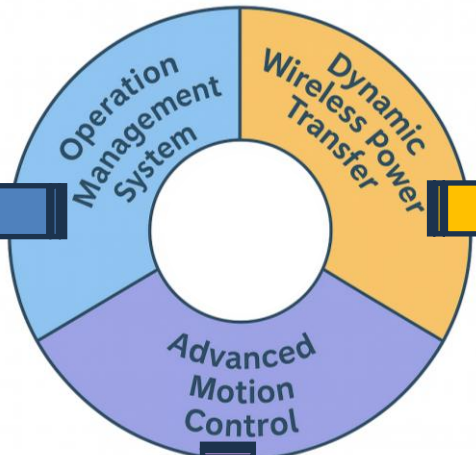


Our work toward the future of e-Mobility

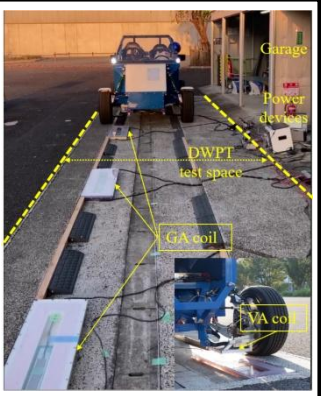


Speed trajectory generation

Torque allocation



Coil design



Position control

T. Koishi et al, "Receiving Side Current-Based Lateral Misalignment Estimation and Automated Steering Control for Dynamic Wireless Power Transfer," IEEE Journal of Emerging and Selected Topics in Industrial Electronics, Vol. 6, No. 1, pp. 41-52 (2024).



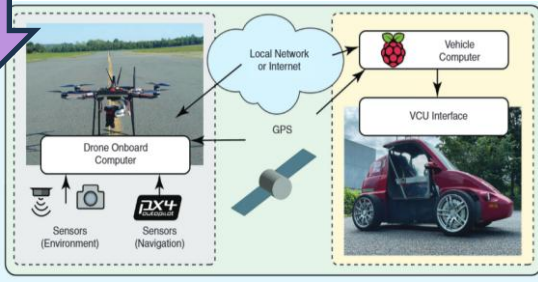
Slip ratio control

Friction force control

B.-M. Nguyen et al, "Slip Control for IWM Vehicles Based on Hierarchical LQR," Control Engineering Practice, Vol. 93, 104179 (2019).

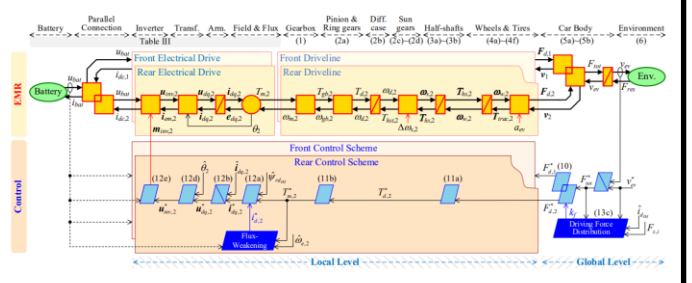


D. Lazzarini et al, "On the Simulation and Experimental Analysis of Imitation Learning for Preview-Based Torque Vectoring Control," Vehicle System Dynamics (2026).



Road condition estimation

V. Beliautsou et al, "Drone-to-Vehicle Integration of Data: Design Concept and Application to Vehicle Automated System," IEEE Vehicular Technology Magazine (2025).



EMR Modeling of Vehicle

A. -T. Nguyen et al, "An Integrated Modeling Framework for Motion Control and Energy Management in Multi-Motor Electric Vehicles," IEEE Open Journal of Vehicular Technology (2025).

Example: Integration of IWM and EPS Control

Without Control



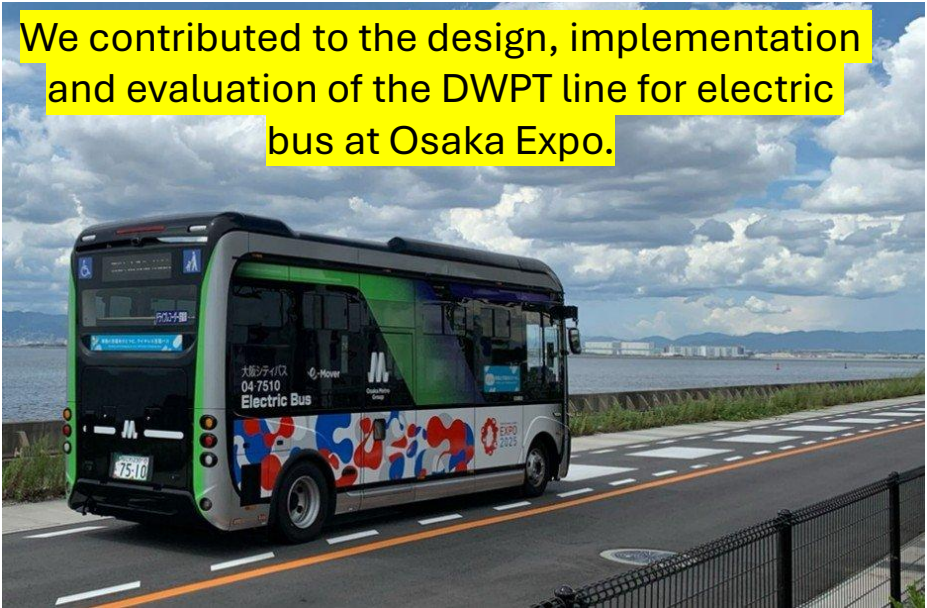
→ Spin on low friction surface
([Click to see the video on Youtube](#))

With Control



→ Stable motion is maintained
([Click to see the video on Youtube](#))

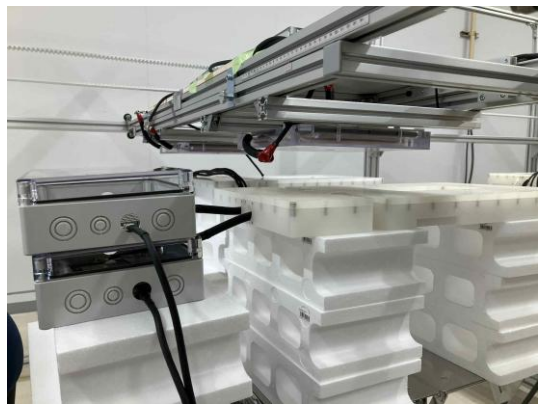
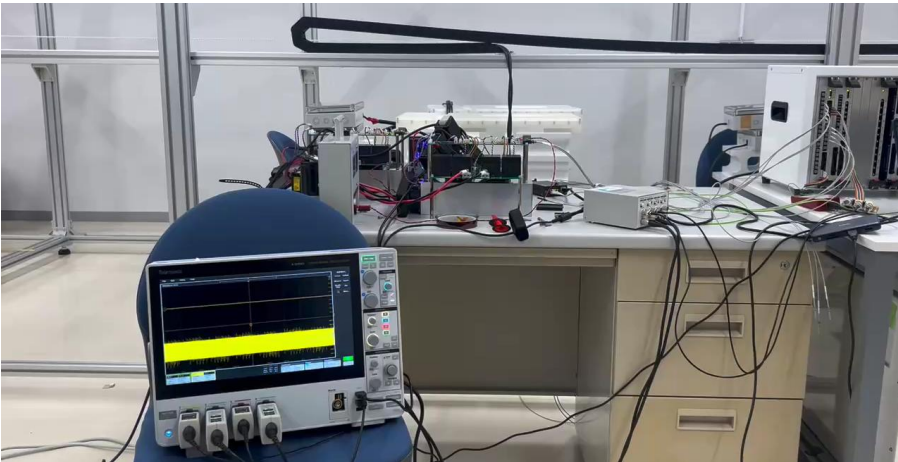
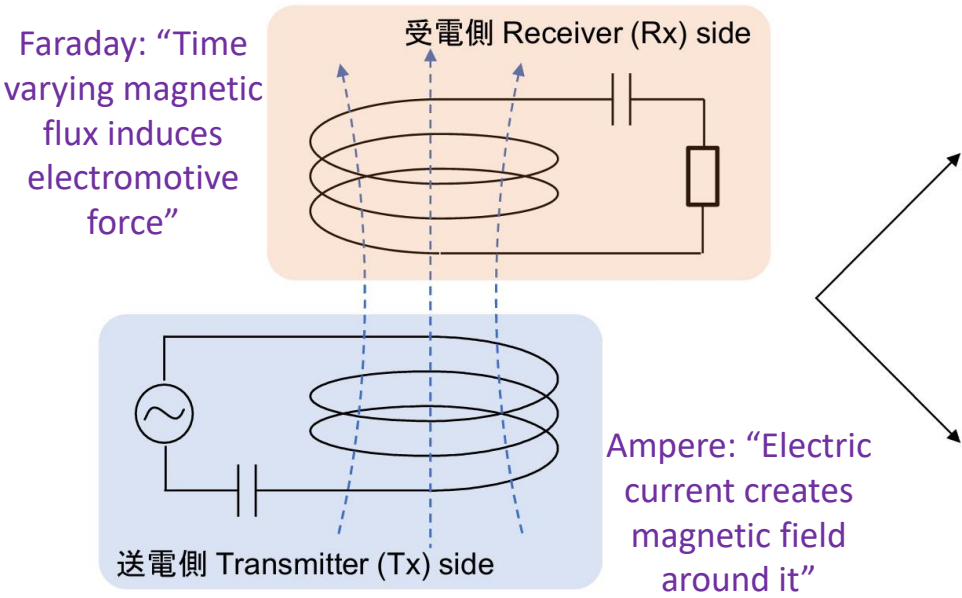
Our DWPT Development



www.k.u-tokyo.ac.jp/information/category/press/11765.html



DWPT section at Kashiwanoha, Kashiwa City, Chiba Prefecture, Japan



Goal of IEEE VTS MVC 2027

Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

- ❖ Enhance the understanding of future e-Mobility concepts, enabling technologies, and key challenges.
- ❖ Demonstrate the need for a *transdisciplinary approach* to developing next-generation e-Mobility systems.
- ❖ Provide participants and the broader community with a multipurpose simulator for vehicle dynamics education, evaluation of research ideas, and collaborative research, etc.
- ❖ Attract participants from diverse academic and professional communities.

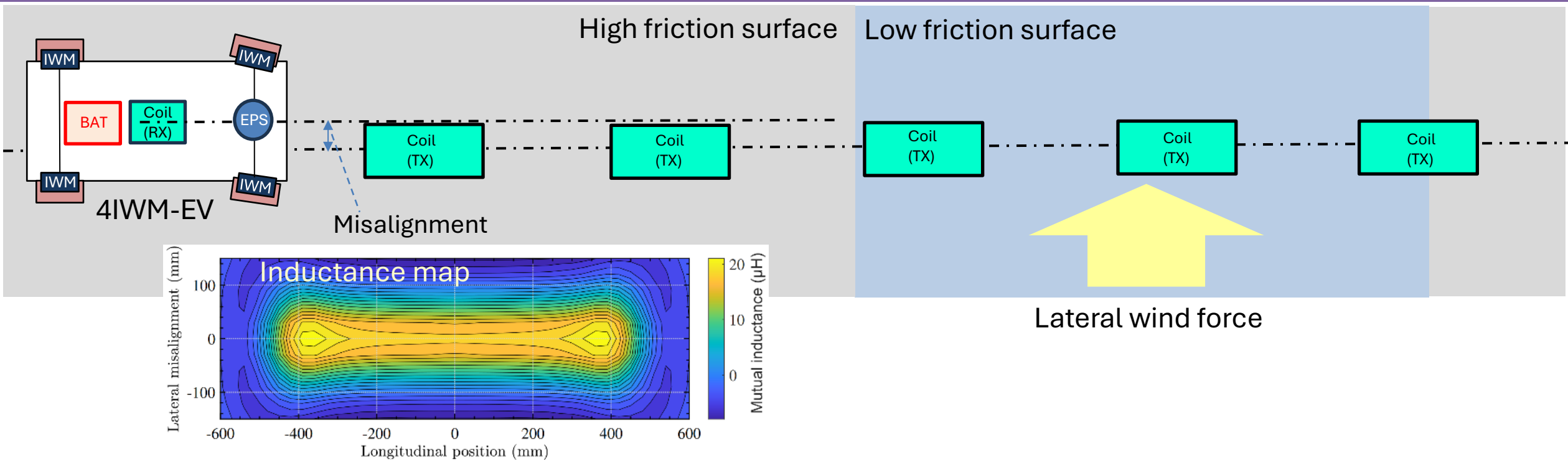
Remark: The simulation is based on a real experimental vehicle operated at the e-Mobility and Control Laboratory, The University of Tokyo. To ensure a feasible problem setting and maintain reasonable computational cost, certain dynamics and components of the real system are simplified or neglected. For example, the dynamics of the energy storage system and the power-electronics control, including coil detection and current control for the DWPT system, are not explicitly modeled. Nevertheless, the resulting model remains sufficiently detailed to provide a challenging simulation environment. The simulator has been developed using MATLAB R2022b, with a sampling time of 0.0002 s.



Agenda

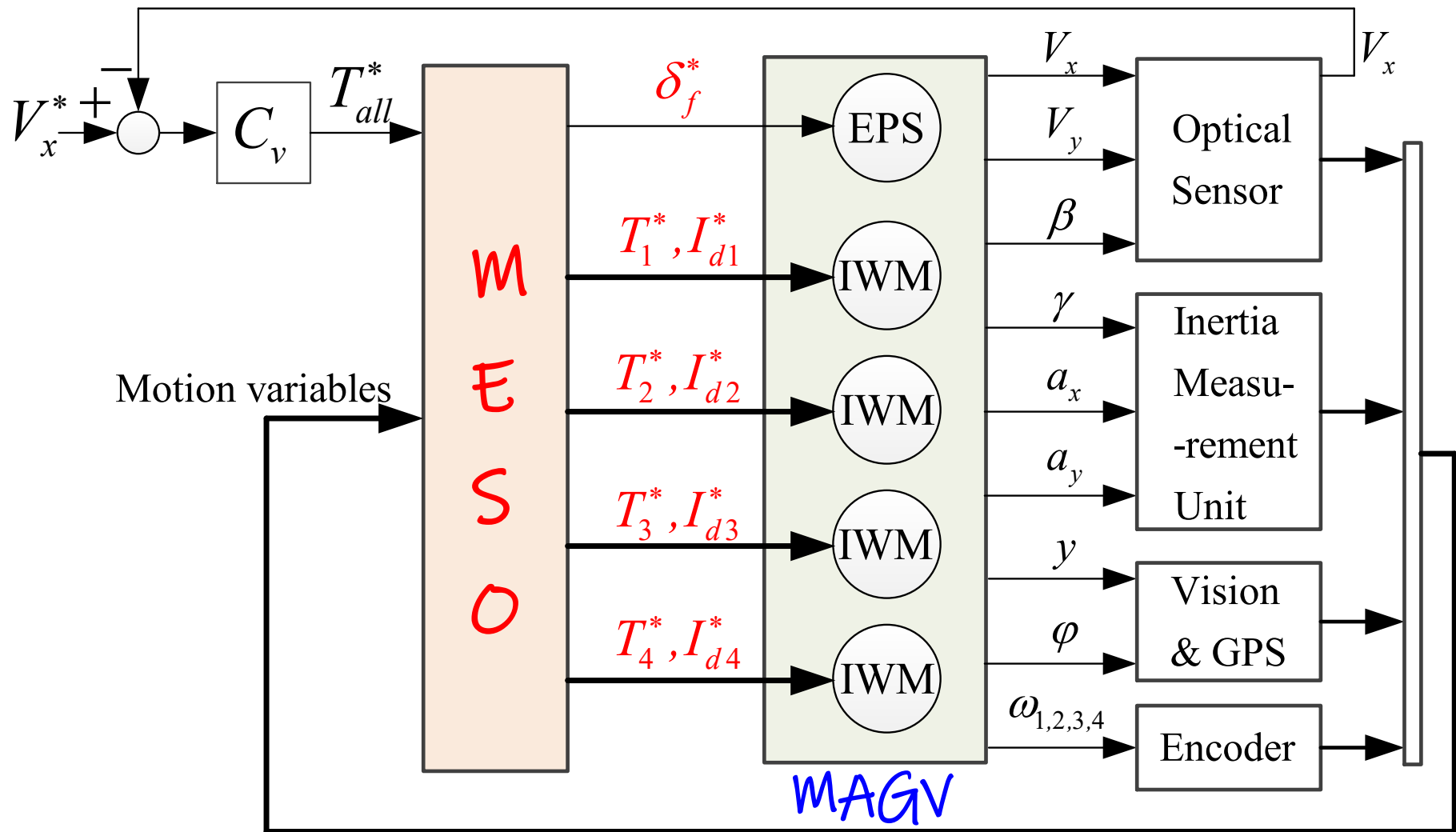
- ❑ Introduction
- ❑ Motivation
- ❑ **Problem setting of MVC2027**
- ❑ MATLAB/Simulink files
- ❑ Demonstration of MESO strategies
- ❑ GitHub repository
- ❑ Registration and Submission

Electric Vehicle and Driving Scenario



- The electric vehicle is actuated by **four** in-wheel-motor and **one** front steering motor. It runs on a *straight* road. The speed pattern and speed controller are given.
- The vehicle may encounter a *low-friction* road surface and strong *lateral wind disturbances*.
- The DWPT is only available if the longitudinal and lateral misalignments are between certain thresholds. For instance, if the lateral misalignment exceeds 70mm, DWPT is shut down.

Block Diagram of MESO



Objectives

The participants are invited to design a strategy that generates {torque distribution ratios, motor d-axis currents, front steering angle} to attain multiple objectives:

(1) Drivetrain efficiency: Minimize the energy consumption:

$$\Phi_1 = \int_0^{t_{end}} P_{all} dt = \int_0^{t_{end}} \sum_{i=1}^4 (P_{m,i} + P_{cu,i} + P_{fe,i}) dt$$

(2) Safe traction: Minimize the wheel slip ratios:

$$\Phi_2 = \int_0^{t_{end}} \sum_{i=1}^4 \lambda_i^2 dt$$

(3) DWPT efficiency: Minimize DWPT efficiency penalty:

$$\Phi_3 = \int_0^{t_{end}} (1 - \eta_{DWPT})^2 dt$$

(4) Driving comfort: Minimize lateral acceleration:

$$\Phi_4 = \int_0^{t_{end}} a_y^2 dt$$

Scoring function: $\Phi = \sigma_1 \Phi_1 + \sigma_2 \Phi_2 + \sigma_3 \Phi_3 + \sigma_4 \Phi_4$

The weighting coefficients account for both the characteristic magnitudes and the relative importance of the four evaluation criteria.

Constraints

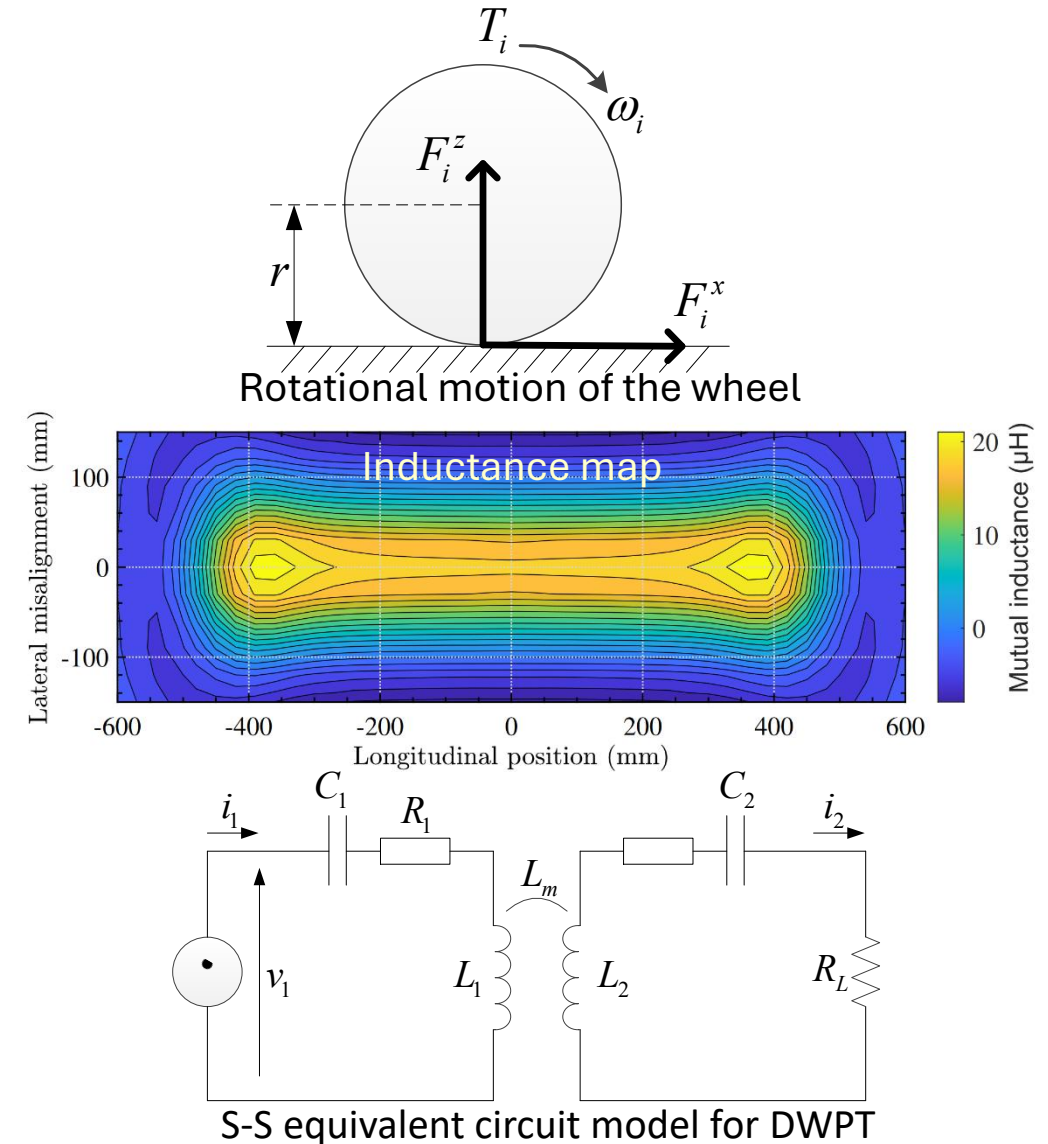
(1) **Total torque command:** $\sum_{i=1}^4 T_i^* = T_{all}^*$, where T_{all}^* is given by the velocity controller.

(2) **Motor torque limits:** $|T_i^*| \leq T_{max,i}, \forall i \in [1,4]$.

(3) **Friction limit circle:** $\sqrt{(F_i^x)^2 + (F_i^y)^2} \leq \mu F_i^z$,

(4) **Coil misalignment:** Wireless power transfer is enabled only when $|\Delta_x| \leq 380$ mm and $|\Delta_y| \leq 70$ mm, where Δ_x and Δ_y denote the longitudinal and lateral misalignments between TX and RX coils, respectively. This is to ensure sufficient mutual inductance and efficiency during power transfer operation:

$$\eta_{DWPT} = \frac{(\omega_0 L_m)^2}{R_1 R_2 + R_1 R_L + (\omega_0 L_m)^2} \frac{R_L}{R_2 + R_L}$$



Agenda

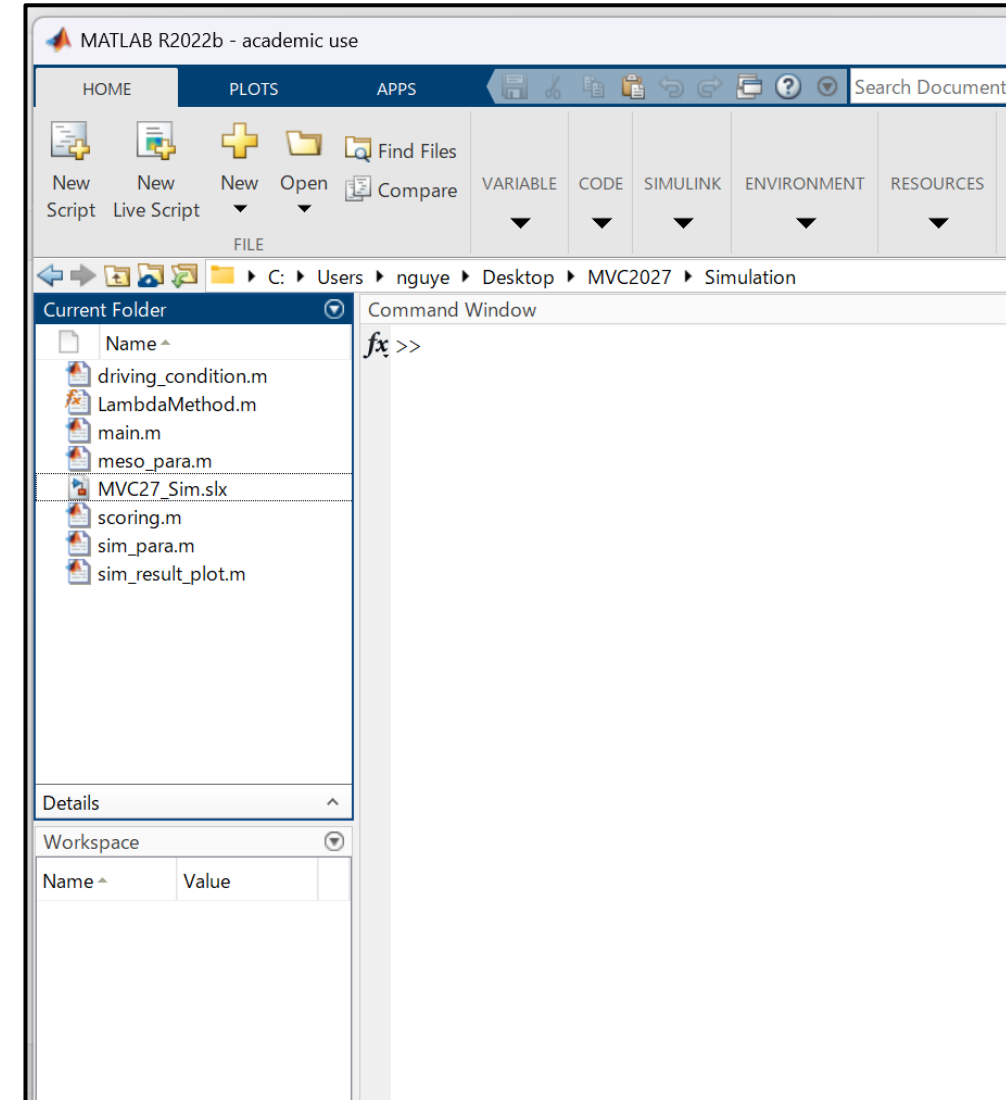
- ❑ Introduction
- ❑ Motivation
- ❑ Problem setting of MVC2027
- ❑ **MATLAB/Simulink files**
- ❑ Demonstration of MESO strategies
- ❑ GitHub repository
- ❑ Registration and Submission

Simulation details

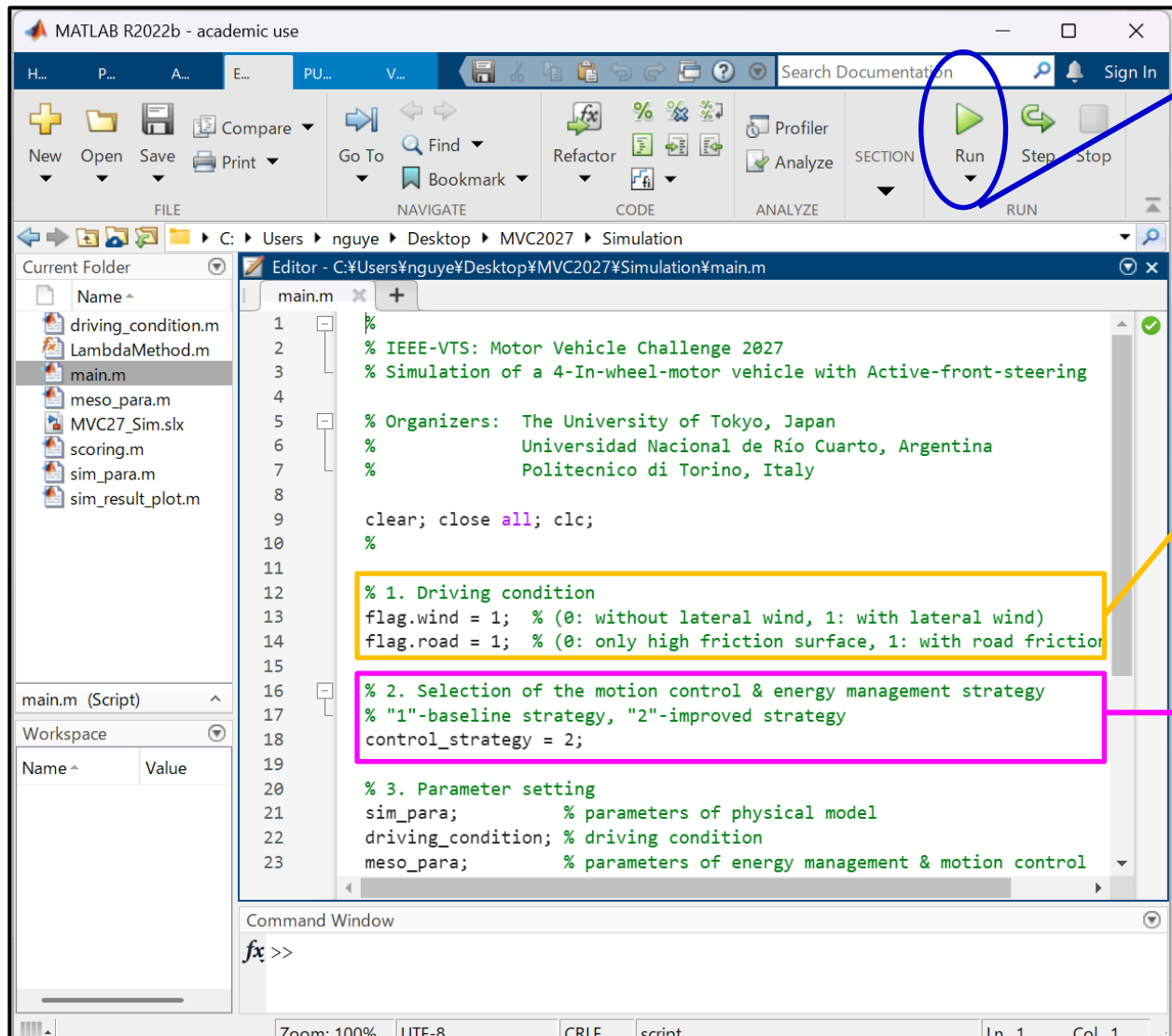
❑ The Simulator has been developed in the MATLAB/Simulink environment (**R2022b**).

❑ Files:

- **main.m** (to run overall simulation)
- **sim_para.m** (parameters of the electric vehicle)
- **meso_para.m** (parameters of the strategies) → *to be designed*
- **driving_condition.m** (setting of road friction & wind force)
- **LambdaMethod.m** (to calculate tire forces using Magic Formula)
- **sim_result_plot.m** (to plot the simulation results)
- **scoring.m** (to calculate the score)
- **MVC27_Sim.slx** (simulation file) → *to be designed*



main.m



❑ “main.m” is used to run the overall simulation.

❑ Before running “main.m”, it is essential to select the driving condition and the motion control & energy management strategy.

❑ For instance, the vehicle runs on the road with lateral wind disturbance and road friction change if we select:
$$\text{flag.wind} = 1; \text{flag.road} = 1;$$

❑ The baseline strategy is selected when
$$\text{control_strategy} = 1;$$

whereas the improved strategy is selected when
$$\text{control_strategy} = 2;$$

sim_para.m

```
1 %
2 % Parameter of the simulator
3
4 % 1. Constant
5 g = 9.81; % acceleration of gravity [m/s^2]
6 pi = 3.14; % number pi
7 eps = 0.05; % small value for anti-division-by-zero i
8 %
9
10 %
11 % 2. Vehicle body parameter
12 Vehicle.M = 880; % total mass [kg]
13 Vehicle.I = 617.0; % yaw moment of inertia [kg.m^2]
14 Vehicle.lf = 0.988; % distance from CG to front axle [m]
15 Vehicle.lr = 0.712; % distance from CG to rear axle [m]
16 Vehicle.l = Vehicle.lf + Vehicle.lr; % wheelbase [m]
17 Vehicle.d = 1.3; % track width
18 Vehicle.df = Vehicle.d; % track width (front) [m]
19 Vehicle.dr = Vehicle.d; % track width (rear) [m]
20 Vehicle.hg = 0.46; % height of CG [ m ]
21 Vehicle.mu0 = 0.0128; % rolling-friction resistance
22 Vehicle.rho = 1.205; % air density
23 Vehicle.Cd = 0.48; % drag coefficient
24 Vehicle.Afpa = 2.4; % frontal projected area [m^2]
25 Vehicle.Nf = (Vehicle.M/2)*g*Vehicle.lr/Vehicle.l; % nominal load (front) [N]
26 Vehicle.Nr = (Vehicle.M/2)*g*Vehicle.lf/Vehicle.l; % nominal load (rear) [N]
```

❑ “sim_para.m” defines the physical parameters of the electric vehicle model.

❑ The parameters are categorized as follows:

% 1. Constant

% 2. Vehicle body parameter

% 3. Wheel parameter

% 4. Steering model

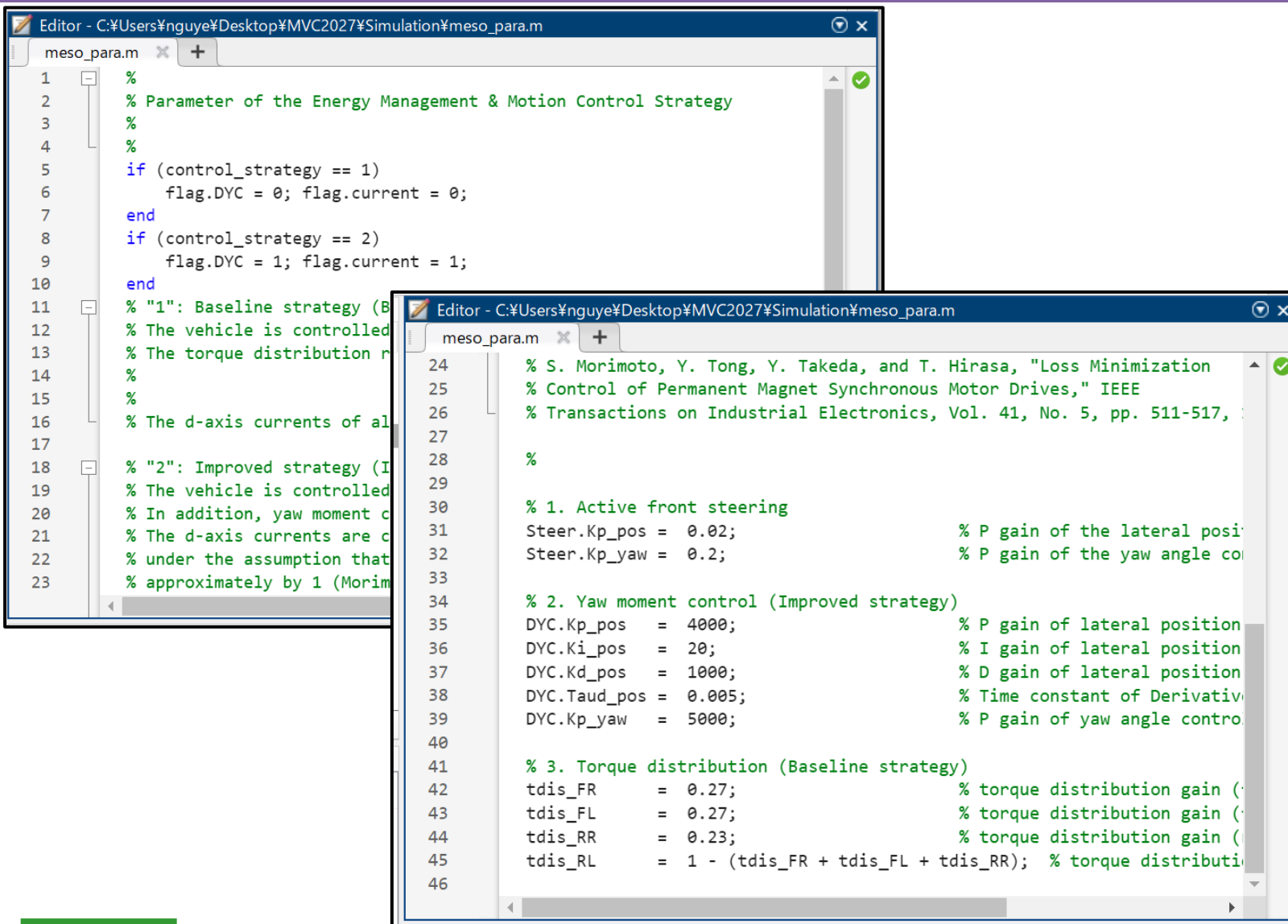
% 5. Road condition

% 6. Motor parameter

% 7. Simulation setting

% 8. Dynamic wireless power transfer

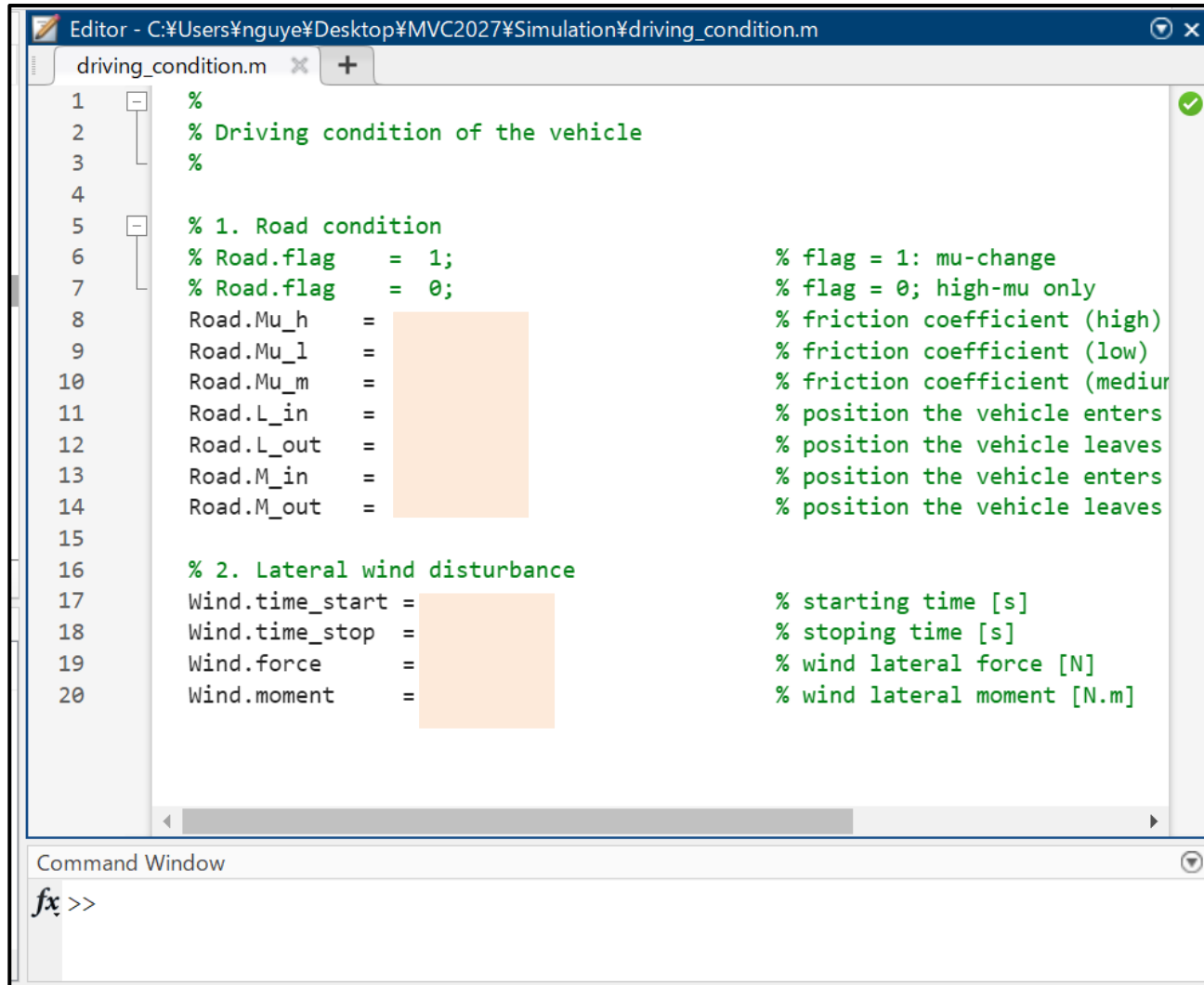
meso_para.m



```
1 %  
2 % Parameter of the Energy Management & Motion Control Strategy  
3 %  
4 %  
5 if (control_strategy == 1)  
6     flag.DYC = 0; flag.current = 0;  
7 end  
8 if (control_strategy == 2)  
9     flag.DYC = 1; flag.current = 1;  
10 end  
11 % "1": Baseline strategy (Baseline)  
12 % The vehicle is controlled by the torque distribution law  
13 % The torque distribution law is used to distribute the torque  
14 %  
15 %  
16 % The d-axis currents of all motors are controlled by the  
17 %  
18 % "2": Improved strategy (Improved)  
19 % The vehicle is controlled by the yaw moment control  
20 % In addition, yaw moment control is used to control the  
21 % The d-axis currents are controlled by the assumption that  
22 % under the assumption that the d-axis currents are controlled  
23 % approximately by 1 (Morimoto, Y. Tong, Y. Takeda, and T. Hirasaka, "Loss Minimization  
24 % S. Morimoto, Y. Tong, Y. Takeda, and T. Hirasaka, "Loss Minimization  
25 % Control of Permanent Magnet Synchronous Motor Drives," IEEE  
26 % Transactions on Industrial Electronics, Vol. 41, No. 5, pp. 511-517, 1994)  
27 %  
28 %  
29 %  
30 % 1. Active front steering  
31 Steer.Kp_pos = 0.02; % P gain of the lateral position  
32 Steer.Kp_yaw = 0.2; % P gain of the yaw angle control  
33  
34 % 2. Yaw moment control (Improved strategy)  
35 DYC.Kp_pos = 4000; % P gain of lateral position  
36 DYC.Ki_pos = 20; % I gain of lateral position  
37 DYC.Kd_pos = 1000; % D gain of lateral position  
38 DYC.Taud_pos = 0.005; % Time constant of Derivative  
39 DYC.Kp_yaw = 5000; % P gain of yaw angle control  
40  
41 % 3. Torque distribution (Baseline strategy)  
42 tdis_FR = 0.27; % torque distribution gain (Front Right)  
43 tdis_FL = 0.27; % torque distribution gain (Front Left)  
44 tdis_RR = 0.23; % torque distribution gain (Rear Right)  
45 tdis_RL = 1 - (tdis_FR + tdis_FL + tdis_RR); % torque distribution  
46
```

- ❑ “meso_para.m” is used to define the parameters of the energy management and motion control strategy.
- ❑ For instance, it consists of the parameters of the following algorithms:
 - ✓ Active front steering control
 - ✓ Yaw moment control
 - ✓ Torque distribution law

driving_condition.m



```
Editor - C:\Users\nguye\Desktop\MVC2027\Simulation\driving_condition.m
driving_condition.m
1 %
2 % Driving condition of the vehicle
3 %
4
5 % 1. Road condition
6 % Road.flag = 1; % flag = 1: mu-change
7 % Road.flag = 0; % flag = 0; high-mu only
8 Road.Mu_h = % friction coefficient (high)
9 Road.Mu_l = % friction coefficient (low)
10 Road.Mu_m = % friction coefficient (medium)
11 Road.L_in = % position the vehicle enters
12 Road.L_out = % position the vehicle leaves
13 Road.M_in = % position the vehicle enters
14 Road.M_out = % position the vehicle leaves
15
16 % 2. Lateral wind disturbance
17 Wind.time_start = % starting time [s]
18 Wind.time_stop = % stoping time [s]
19 Wind.force = % wind lateral force [N]
20 Wind.moment = % wind lateral moment [N.m]

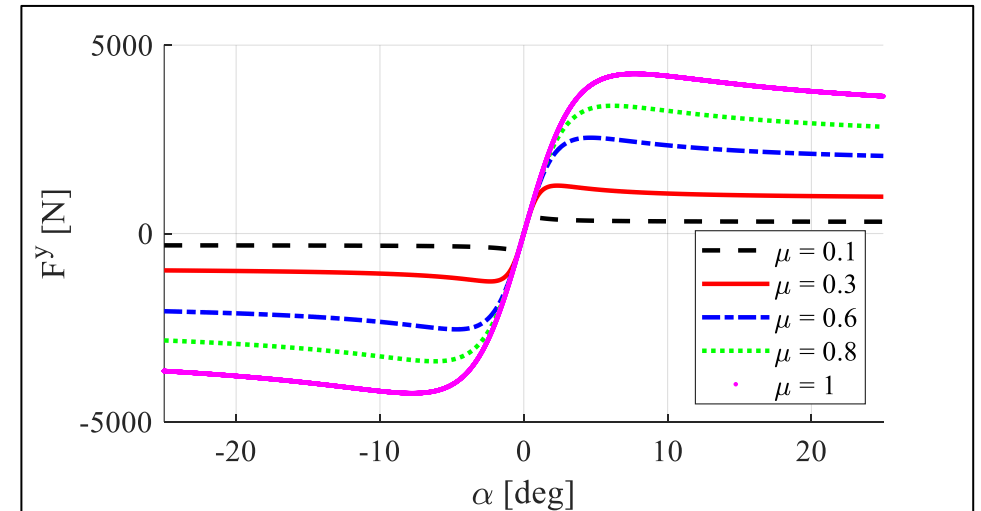
Command Window
fx >>
```

- ❑ “driving_condition.m” is used to define the road conditions and specify the lateral wind force disturbance.
- ❑ The control strategy must not directly access information and parameters defined by “driving_condition.m.”

LambdaMethod.m

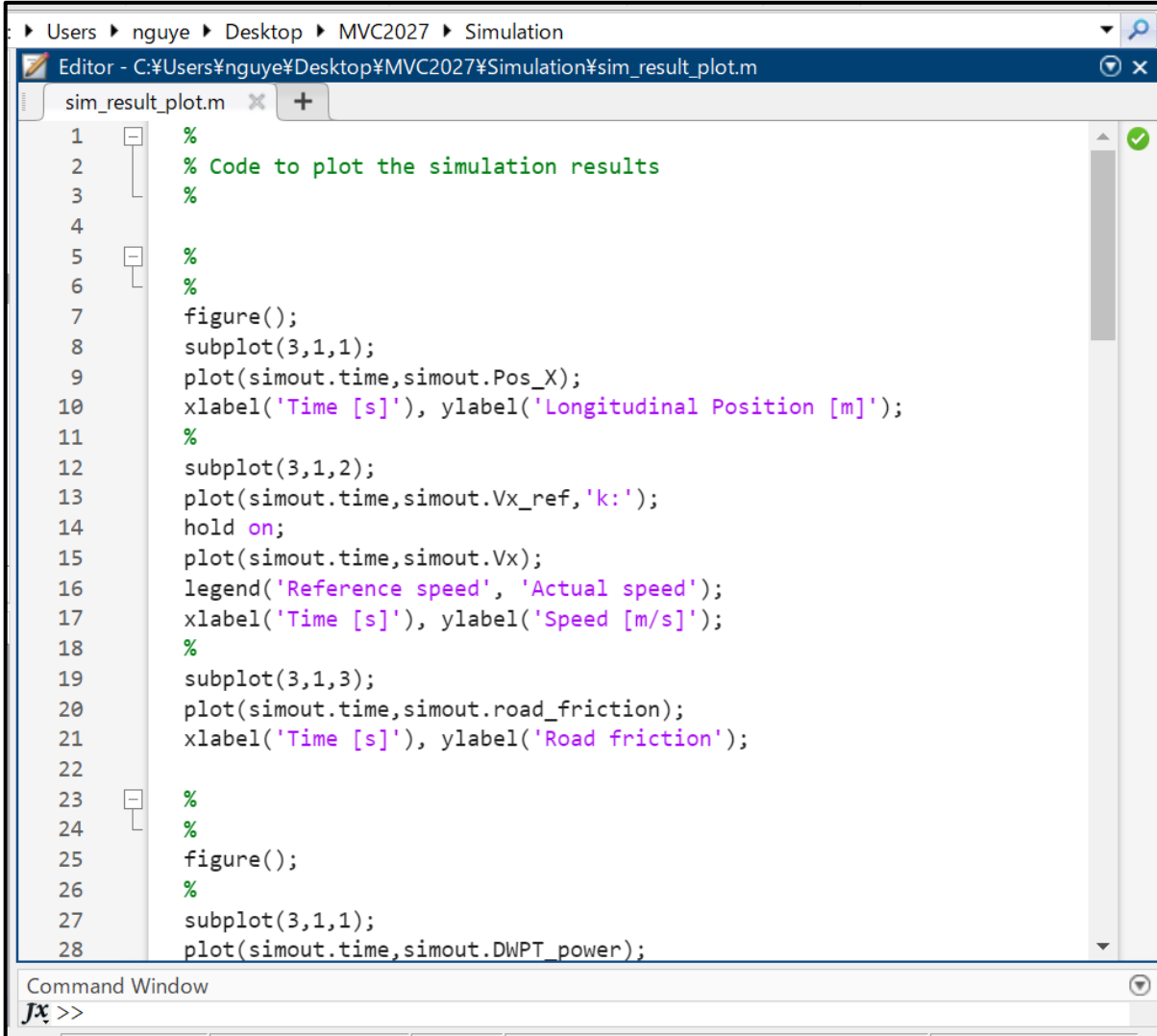
```
C:\Users\nguye\Desktop\MVC2027\Simulation
Editor - C:\Users\nguye\Desktop\MVC2027\Simulation\LambdaMethod.m
LambdaMethod.m
1 function [sys,x0,str,ts] = sfuntmpl(t,x,u,flag)
2 %SFUNTMPL General M-file S-function template
3 % 2D Tire Model Based on Lambda-Method.
4 % This S-Function is to calculate the
5 % Longitudinal (Driving) Force Fx
6 % and Lateral Force Fy
7 %
8 % based on the Magic Formula, proposed by H. B. Pacejka
9 % in "Tire and Vehicle Dynamics,"
10 % Elsevier Science, 2005.
11 %
12 % and based on the "Lambda Method", proposed by Y. Horiuchi.
13 % in "A Proposition of the Simple Tire Model for the Vehicle Stability
14 % Assist System," Proceedings of Spring Congress
15 % of Society of Automotive Engineers of Japan, No. 64-98 (1998).
16 %
17 switch flag,
18
19 %%%%%%%%%%%%%%%
20 % Initialization %
21 %%%%%%%%%%%%%%%
22 case 0,
23     [sys,x0,str,ts]=mdlInitializeSizes;
24
25 %%%%%%%%%%%%%%%
26 % Derivatives %
27 %%%%%%%%%%%%%%%
28 case 1,
```

□ “**LambdaMethod.m**” is an S-Function that calculates the longitudinal and lateral forces at each wheel using Pacejka’s Magic Formula.



Example of lateral tire force characteristic

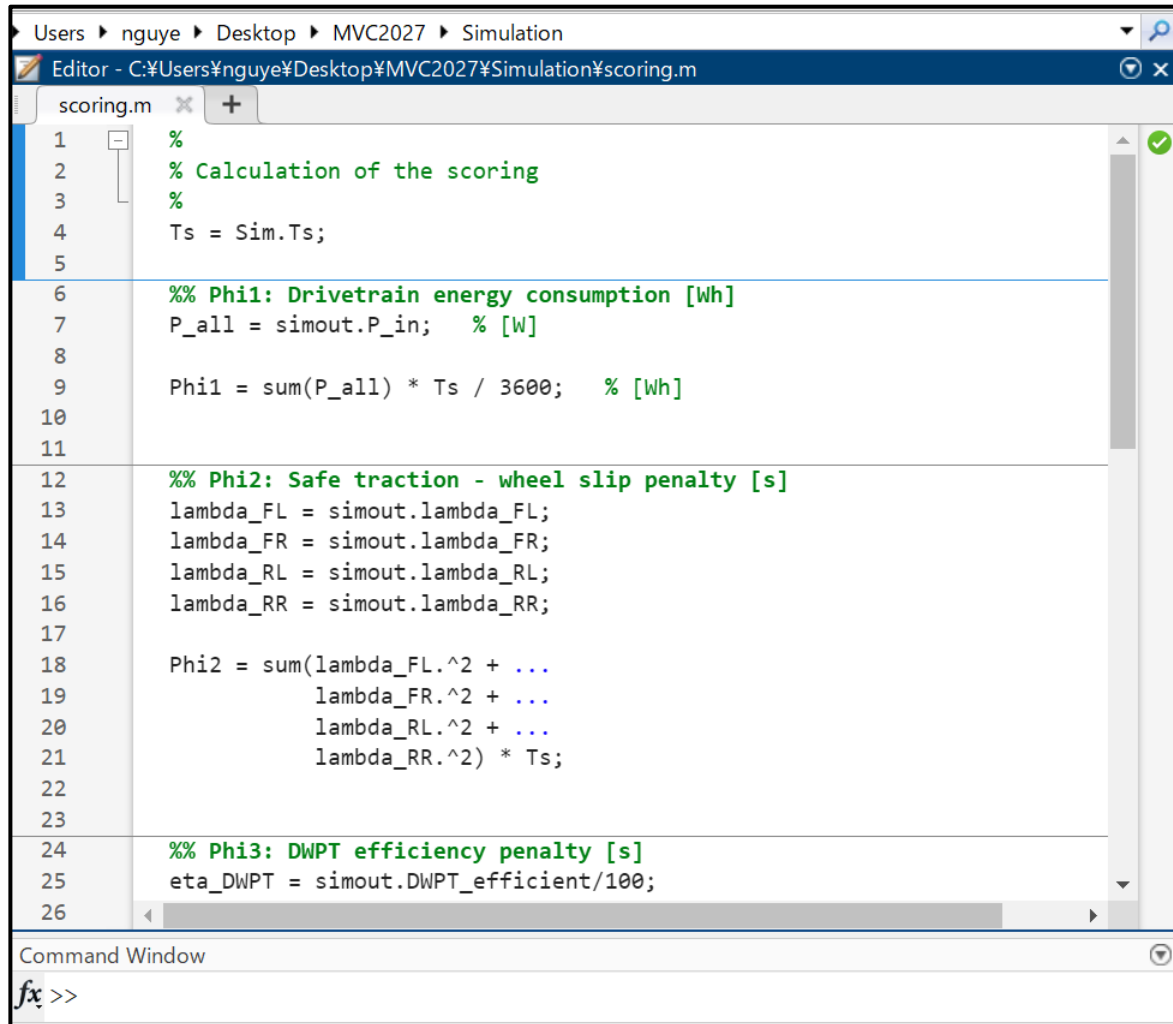
sim_result_plot.m



```
1 %  
2 % Code to plot the simulation results  
3 %  
4  
5 %  
6 %  
7 figure();  
8 subplot(3,1,1);  
9 plot(simout.time,simout.Pos_X);  
10 xlabel('Time [s]'), ylabel('Longitudinal Position [m]');  
11 %  
12 subplot(3,1,2);  
13 plot(simout.time,simout.Vx_ref,'k:');  
14 hold on;  
15 plot(simout.time,simout.Vx);  
16 legend('Reference speed', 'Actual speed');  
17 xlabel('Time [s]'), ylabel('Speed [m/s]');  
18 %  
19 subplot(3,1,3);  
20 plot(simout.time,simout.road_friction);  
21 xlabel('Time [s]'), ylabel('Road friction');  
22  
23 %  
24 %  
25 figure();  
26 %  
27 subplot(3,1,1);  
28 plot(simout.time,simout.DWPT_power);
```

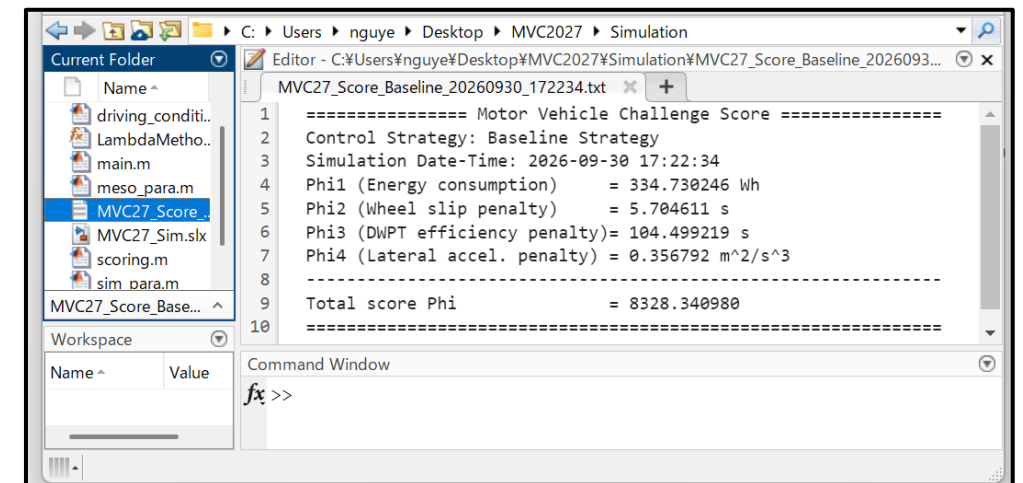
- ❑ “**sim_result_plot.m**” is to plot the main simulation results, such as
 - ✓ vehicle velocity
 - ✓ road friction coefficient
 - ✓ wheel velocities
 - ✓ lateral acceleration
 - ✓ yaw rate
 - ✓ lateral misalignment
 - ✓ efficiency of DWPT
 - ✓ energy consumed by motors
 - ✓ motor d-axis currents
 - ✓ motor torques
 - ✓ front steering angle

scoring.m



```
1 %  
2 % Calculation of the scoring  
3 %  
4 Ts = Sim.Ts;  
5  
6 %% Phi1: Drivetrain energy consumption [Wh]  
7 P_all = simout.P_in; % [W]  
8  
9 Phi1 = sum(P_all) * Ts / 3600; % [Wh]  
10  
11  
12 %% Phi2: Safe traction - wheel slip penalty [s]  
13 lambda_FL = simout.lambda_FL;  
14 lambda_FR = simout.lambda_FR;  
15 lambda_RL = simout.lambda_RL;  
16 lambda_RR = simout.lambda_RR;  
17  
18 Phi2 = sum(lambda_FL.^2 + ...  
19             lambda_FR.^2 + ...  
20             lambda_RL.^2 + ...  
21             lambda_RR.^2) * Ts;  
22  
23  
24 %% Phi3: DWPT efficiency penalty [s]  
25 eta_DWPT = simout.DWPT_efficient/100;  
26
```

- ❑ “scoring.m” is used to calculate the score and display the results in the MATLAB Command Window.
- ❑ The score is saved in a text file named `MVC27_Score_StrategyName_Date_Time.txt`
 - StrategyName is Baseline or Improved.
 - Date_Time indicates the date and time at which the simulation was executed.

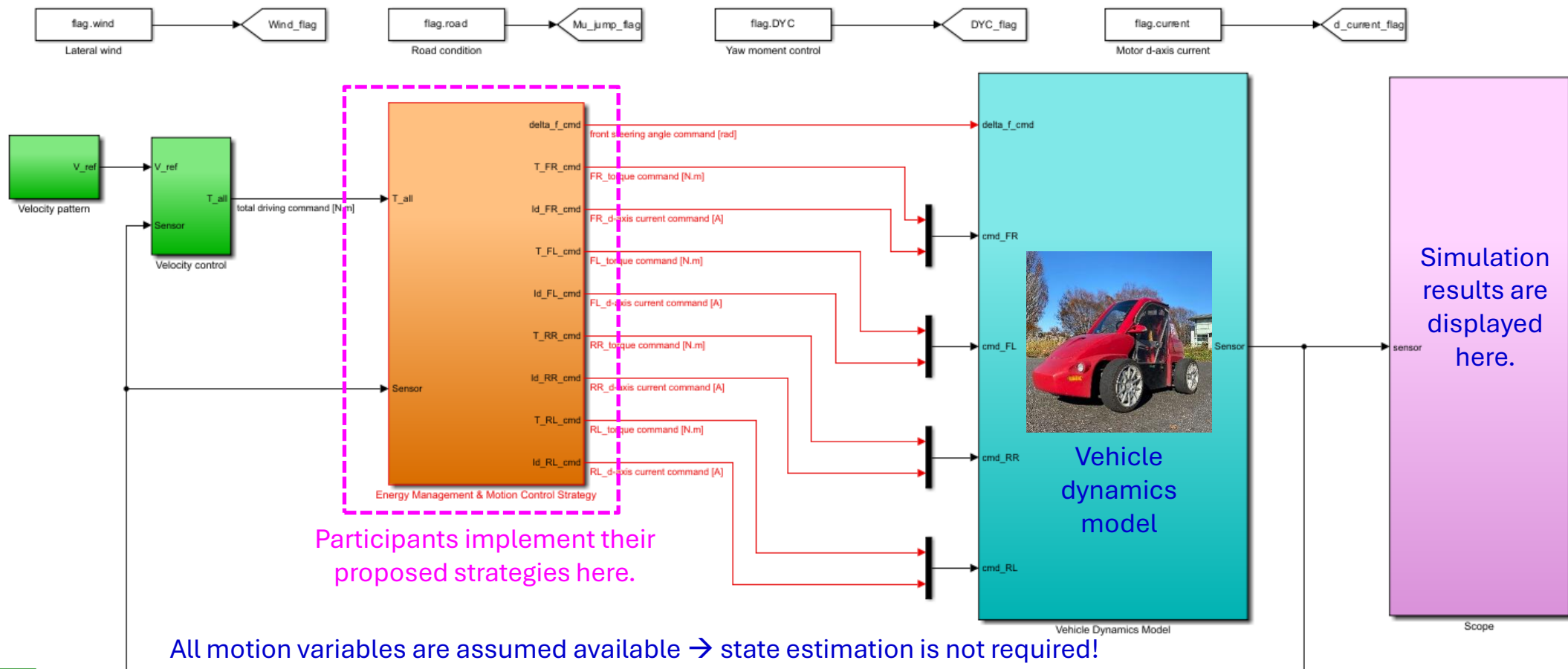


```
1 ===== Motor Vehicle Challenge Score =====  
2 Control Strategy: Baseline Strategy  
3 Simulation Date-Time: 2026-09-30 17:22:34  
4 Phi1 (Energy consumption) = 334.730246 Wh  
5 Phi2 (Wheel slip penalty) = 5.704611 s  
6 Phi3 (DWPT efficiency penalty)= 104.499219 s  
7 Phi4 (Lateral accel. penalty) = 0.356792 m^2/s^3  
8 -----  
9 Total score Phi = 8328.340980  
10 =====
```

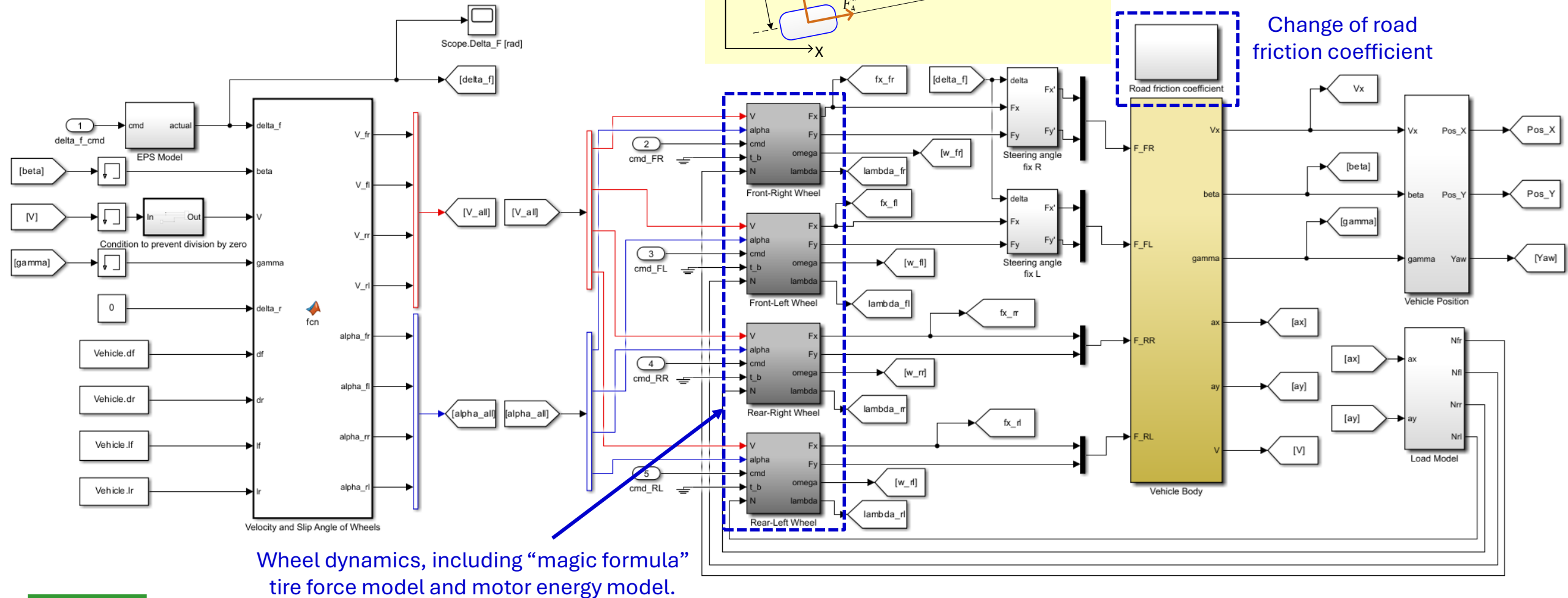
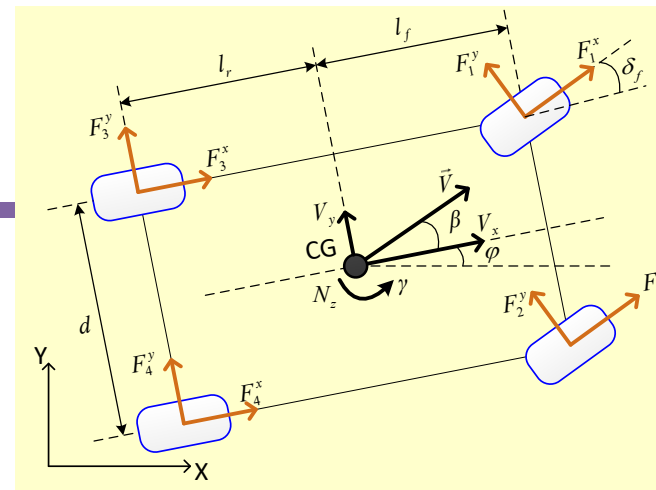
Outline of MVC27_Sim.slx

Energy Management and Motion Control of an Electric Vehicle driven by Four In-Wheel-Motors with Active Front Steering on Dynamic Wireless Power Transfer Road

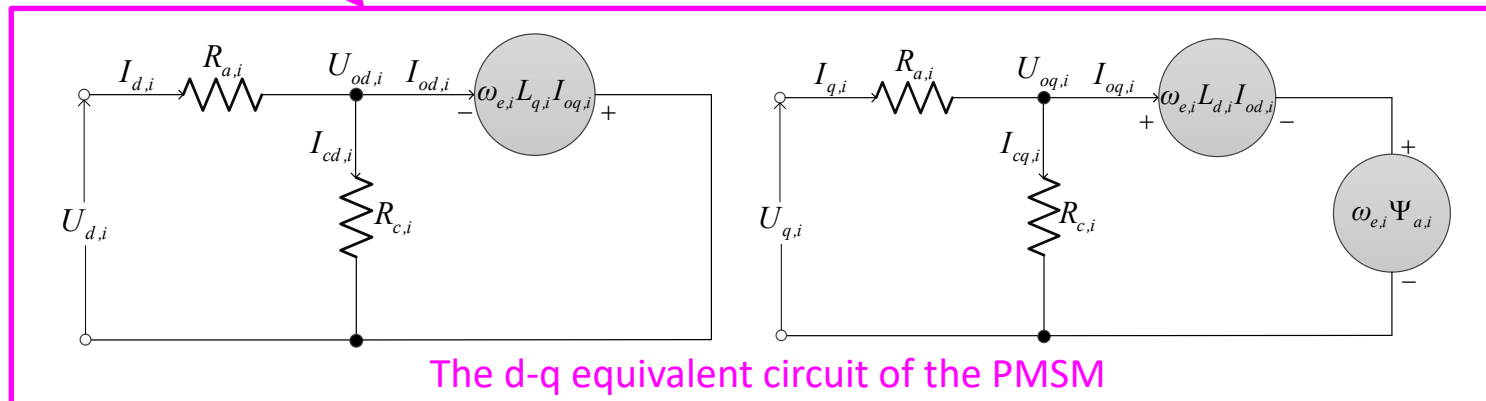
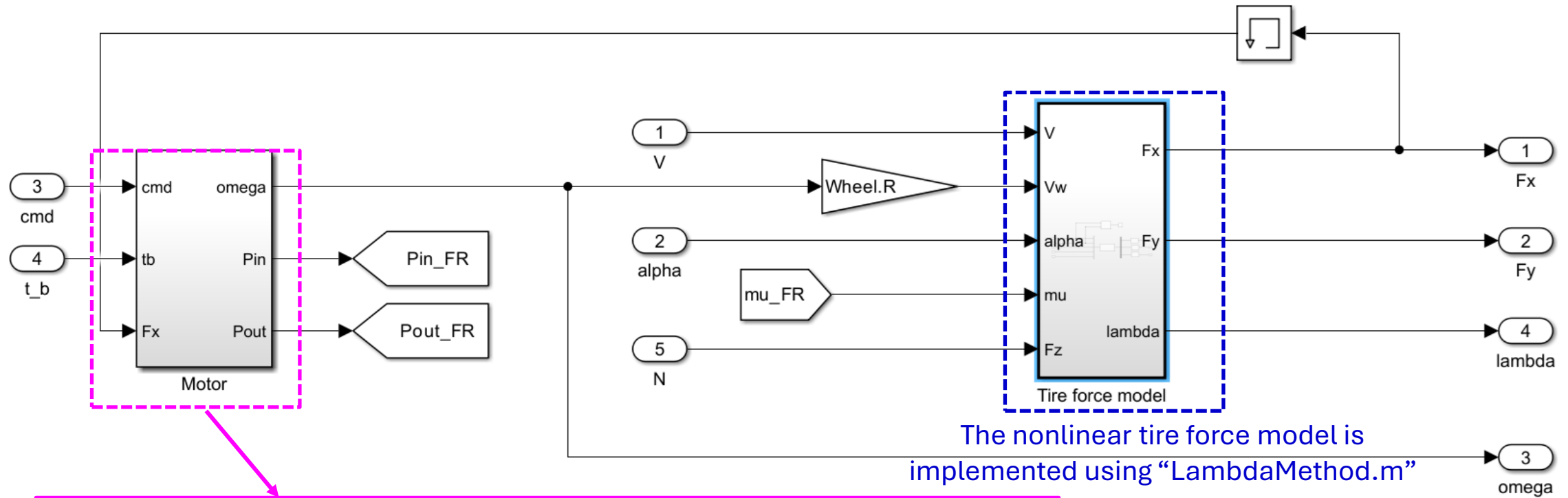
(The flags are used to set up the simulation condition)



Vehicle dynamics model



Wheel dynamics



Agenda

- ❑ Introduction
- ❑ Motivation
- ❑ Problem setting of MVC2027
- ❑ MATLAB/Simulink files
- ❑ **Demonstration of MESO strategies**
- ❑ GitHub repository
- ❑ Registration and Submission

Examples of MESO strategies

Baseline strategy (BS):

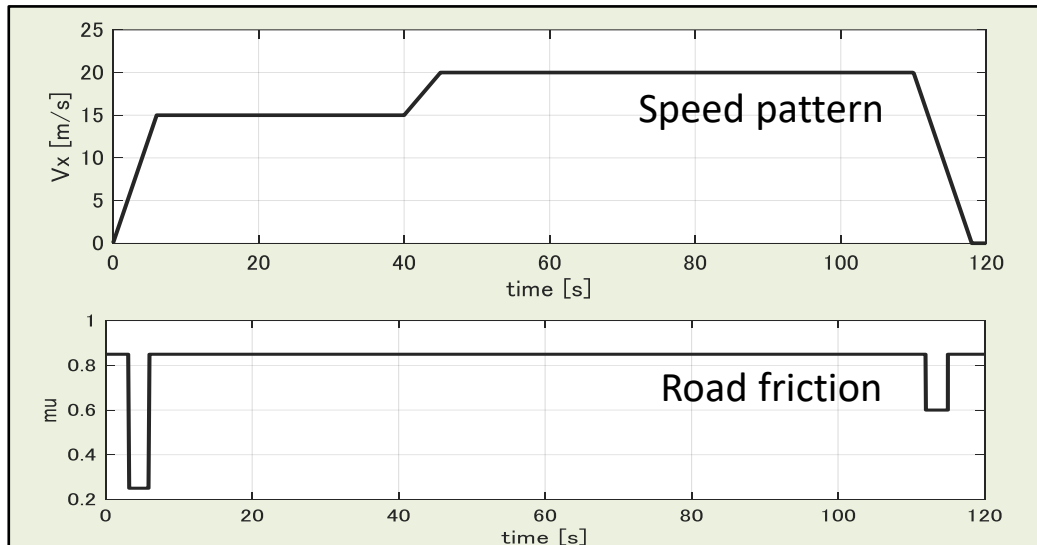
❖ Front EPS angle control:

$$\delta_f^* = K_{\delta y, p} e_y + K_{\delta \varphi, p} e_\varphi$$

where $e_y = y^* - y$, $e_\varphi = \varphi^* - \varphi$ with the reference values $y^* = 0$ and $\varphi^* = 0$. The control gains are $K_{\delta y, p}$ and $K_{\delta \varphi, p}$.

❖ IWM torque distribution: The torque distribution ratios are fixed as: $\{t_{1,2} = 0.27; t_{3,4} = 0.23\} \rightarrow$ no yaw moment is applied.

❖ Motor control: The d-axis current is $I_{d,i}^* = 0$ for all IWMs.



Improved strategy (IS):

❖ Front EPS angle control: The same as that of the BS.

❖ IWM torque distribution: The IWM torques are distributed according to a yaw moment control law:

$$\begin{cases} T_{1,3}^* = \frac{T_{all}^*}{4} - \frac{r}{2d} N^{z*}; & T_{2,4}^* = \frac{T_{all}^*}{4} + \frac{r}{2d} N^{z*} \\ N^{z*} = K_{Ny, p} e_y + K_{Ny, i} \int_0^t e_y dt + K_{Ny, d} \frac{de_y}{dt} + K_{N\varphi, p} e_\varphi \end{cases}$$

with the control gains $K_{Ny, p}$, $K_{Ny, i}$, $K_{Ny, d}$, and $K_{N\varphi, p}$.

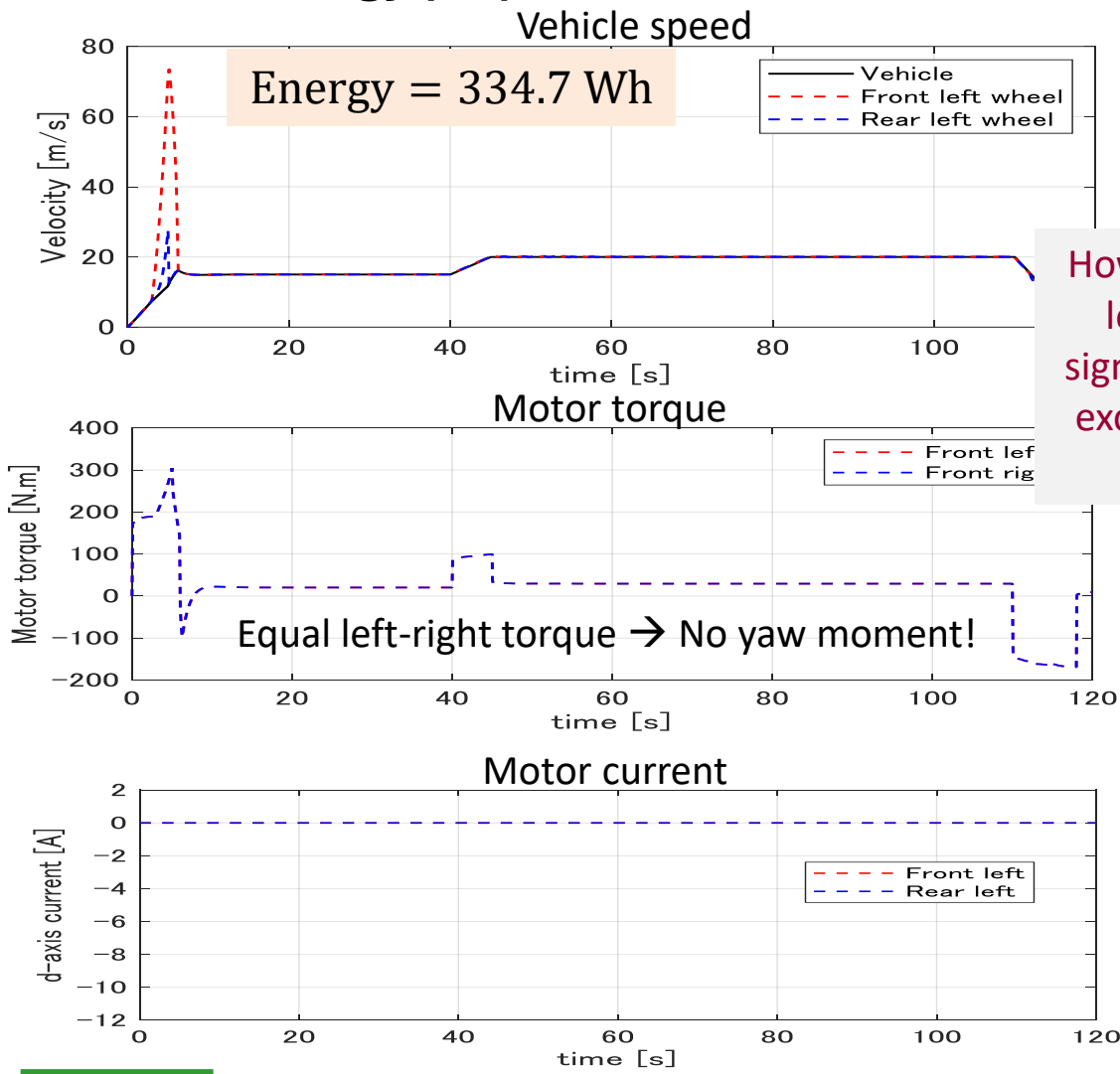
❖ Motor control: Assuming that the motor salient coefficients are close to unity, the d-axis current that minimizes motor losses is given by the following formula (Morimoto *et al*, 1994):

$$I_{od, i}^* = - \frac{\omega_{e, i}^2 L_{d, i} (R_{a, i} + R_{c, i}) \Psi_{a, i}}{R_{a, i} R_{c, i}^2 + \omega_{e, i}^2 L_{d, i}^2 (R_{a, i} + R_{c, i})}$$

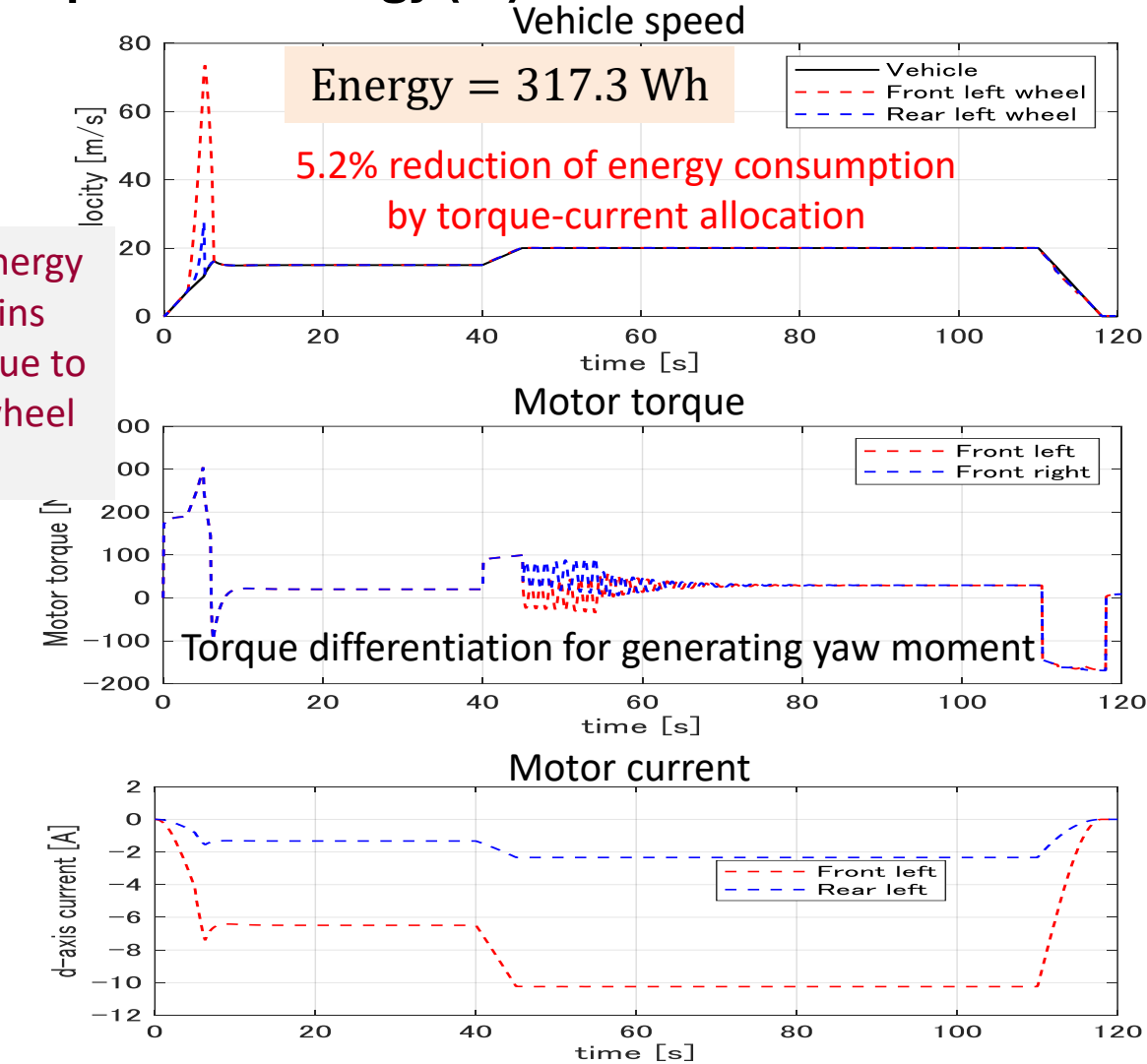
S. Morimoto, Y. Tong, Y. Takeda and T. Hirasa, "Loss minimization control of permanent magnet synchronous motor drives," in *IEEE Transactions on Industrial Electronics*, vol. 41, no. 5, pp. 511-517, Oct. 1994, doi: 10.1109/41.315269.

Simulation results

Baseline strategy (BS)



Improved strategy (IS)



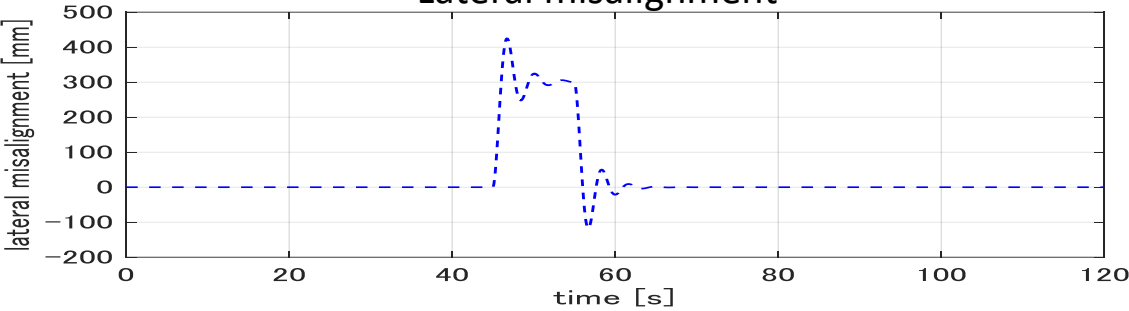
However, energy loss remains significant due to excessive wheel slip.

5.2% reduction of energy consumption by torque-current allocation

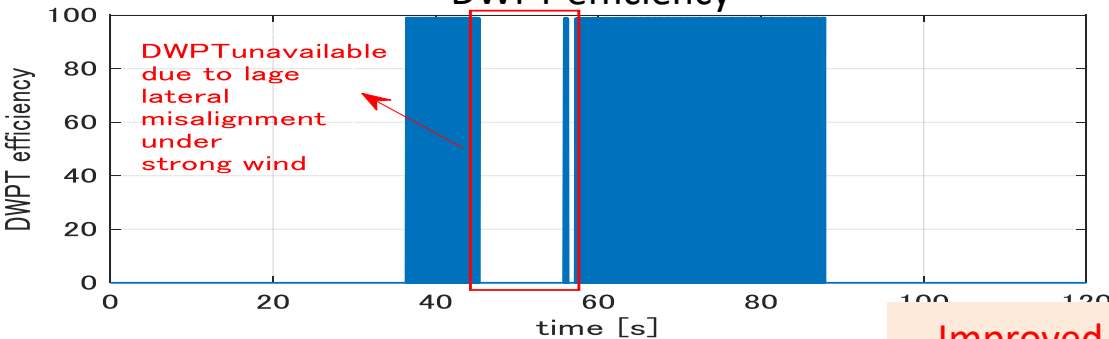
Simulation results

Baseline strategy (BS)

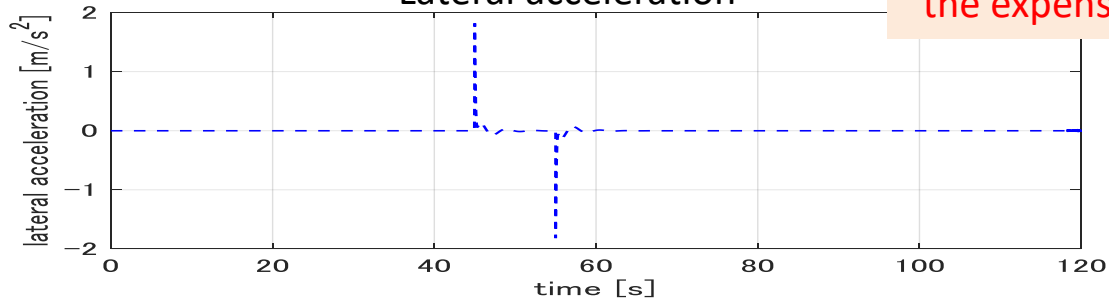
Lateral misalignment



DWPT efficiency

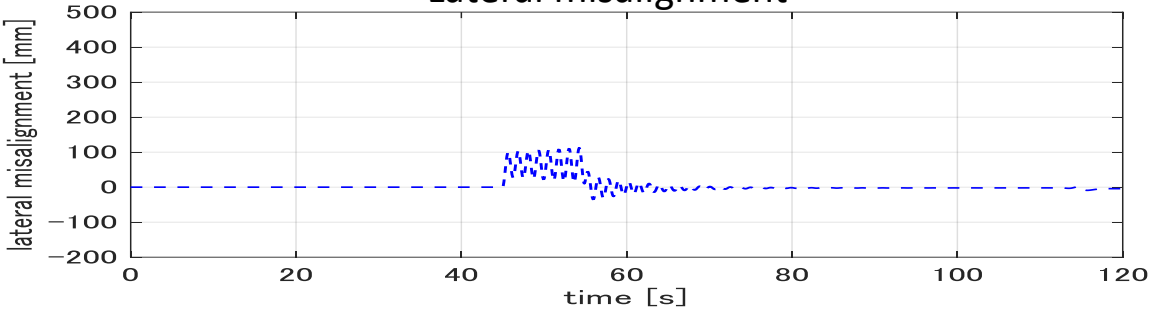


Lateral acceleration

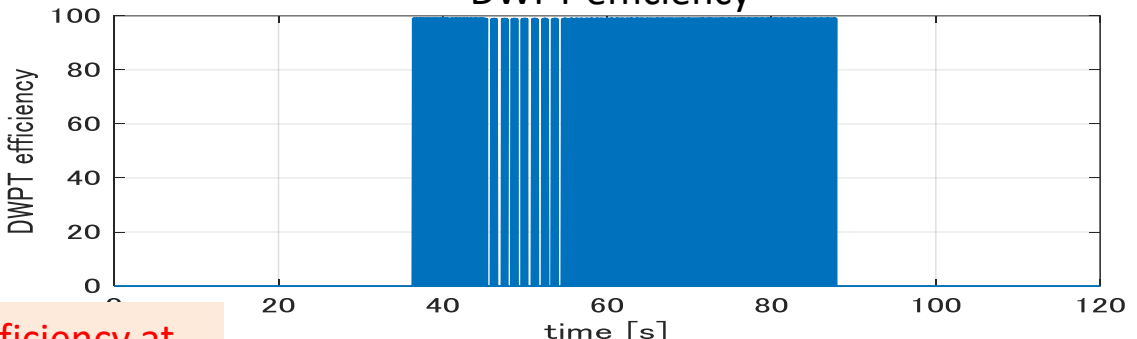


Improved strategy (IS)

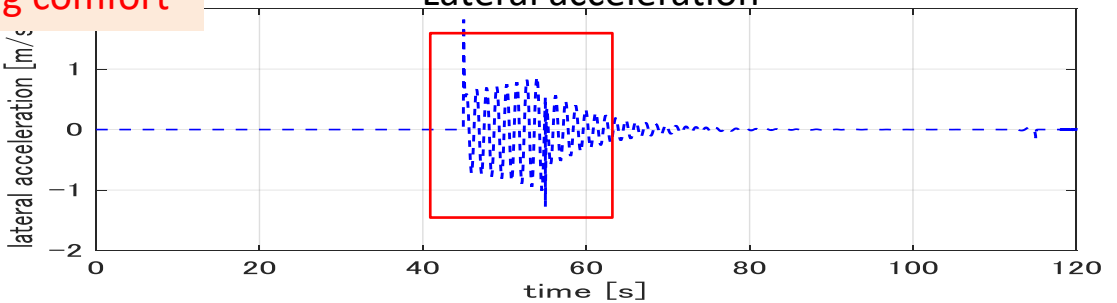
Lateral misalignment



DWPT efficiency



Lateral acceleration



Improved DWPT efficiency at the expense of driving comfort

Scoring of MESO Strategies

Baseline strategy (BS) $\phi = 994.072497$

The screenshot shows the MATLAB R2022b interface with the 'scoring.m' script open. The script contains comments for '2. Selection of the motion control & energy management' and '3. Parameter setting'. The Command Window displays the 'Motor Vehicle Challenge Score' for the Baseline Strategy, including simulation date-time, individual penalties (Phi1 to Phi4), and the total score Phi.

```
==== Motor Vehicle Challenge Score =====
Control Strategy: Baseline Strategy
Simulation Date-Time: 2026-10-04 19:41:28
Phi1 (Energy consumption)    = 334.730246 Wh
Phi2 (Wheel slip penalty)    = 5.704611 s
Phi3 (DWPT efficiency penalty)= 104.499219 s
Phi4 (Lateral accel. penalty) = 0.356792 m^2/s^3
-----
Total score Phi              = 994.072497
=====
Score file saved as: MVC27_Score_Baseline_20261004_194128.txt
fx >>
```

Improved strategy (IS) $\phi = 968.530144$

The screenshot shows the MATLAB R2022b interface with the 'scoring.m' script open. The script contains comments for '2. Selection of the motion control & energy management' and '3. Parameter setting'. The Command Window displays the 'Motor Vehicle Challenge Score' for the Improved Strategy, including simulation date-time, individual penalties (Phi1 to Phi4), and the total score Phi.

```
==== Motor Vehicle Challenge Score =====
Control Strategy: Improved Strategy
Simulation Date-Time: 2026-10-04 20:00:04
Phi1 (Energy consumption)    = 317.270407 Wh
Phi2 (Wheel slip penalty)    = 5.395927 s
Phi3 (DWPT efficiency penalty)= 102.188848 s
Phi4 (Lateral accel. penalty) = 2.093757 m^2/s^3
-----
Total score Phi              = 968.530144
=====
Score file saved as: MVC27_Score_Improved_20261004_200004.txt
fx >>
```

Score Analysis

Metric	Baseline Strategy (BS)	Improved Strategy (IS)	Change (IS – BS)	Change
Φ_1 : Energy consumption	334.730246 Wh	317.270407 Wh	–17.459839 Wh	–5.22%
Φ_2 : Wheel slip penalty	5.704611 s	5.395927 s	–0.308684 s	–5.41%
Φ_3 : DWPT efficiency penalty	104.499219 s	102.188848 s	–2.310371 s	–2.21%
Φ_4 : Lateral accel. penalty	0.356792 m ² /s ³	2.093757 m ² /s ³	+1.736965 m ² /s ³	+486.83%
Total score Φ	994.072497	968.530144	–25.542353	–2.57%

The improved strategy reduces the total score by **2.57%**, mainly through reductions in energy consumption, wheel slip, and DWPT efficiency penalty, while sacrificing lateral-acceleration performance.

Agenda

- ❑ Introduction
- ❑ Motivation
- ❑ Problem setting of MVC2027
- ❑ MATLAB/Simulink files
- ❑ Demonstration of MESO strategies
- ❑ **GitHub repository**
- ❑ Registration and Submission

GitHub Repository

It contains:

- Introduction to MVC2027
- Simulation files
- Deadlines
- Competition rules
- Registration link
- Submission link
- Any other relevant information/materials
 - MVC2027 paper
 - paper presentation slide
 - workshop slide

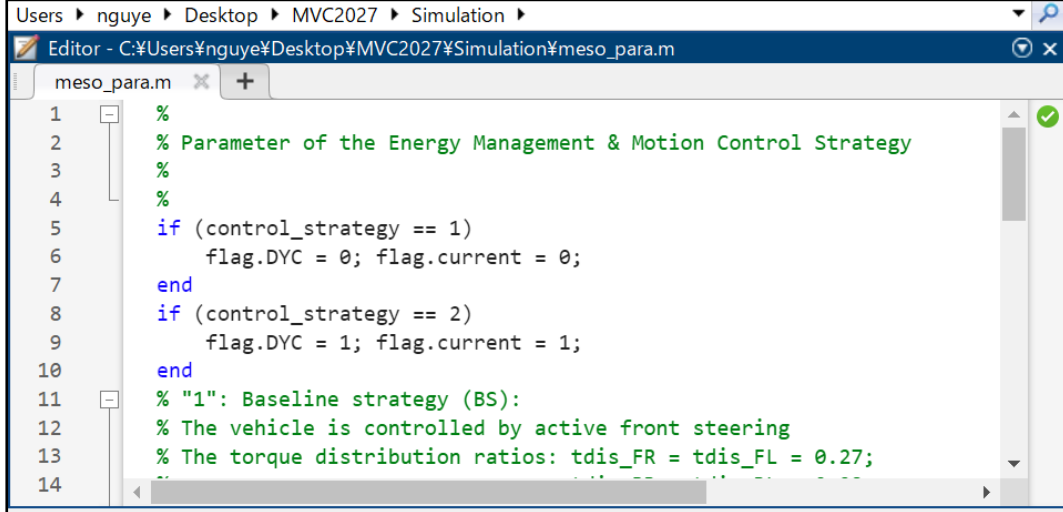
https://github.com/VTsociety/MVC_2027/



- Registration deadline: 31 January 2027
- Submission: 1 March 2027
- Results: 31 March 2027

Competition Rules

1. The MATLAB version used for the submitted files must be **R2022b**.
2. Participants are allowed to modify only the “**meso_para.m**” script and the “**Energy Management & Motion Control Strategy**” block provided in the simulation file.
3. All parameters in “**sim_para.m**”, as well as all measurements available from the output of the “Vehicle Dynamics Model” block, may be used to develop the proposed strategy.
4. All other scripts and simulation blocks **must not** be modified. Any modifications made to scripts other than “**meso_para.m**” or to subsystems other than “**Energy Management & Motion Control Strategy**” will be discarded during the evaluation.
5. The final score will be calculated as the average of the scores obtained using the example and secret driving cycles.
6. A proposed strategy will be disqualified if the vehicle *fails to complete* either driving cycle (e.g., the vehicle stops before reaching the end of the driving cycle), or if the *lateral position of the vehicle exceeds 5 m* at any time during the simulation.



The screenshot shows a MATLAB editor window titled "Editor - C:\Users\nguye\Desktop\MVC2027\Simulation\meso_para.m". The script content is as follows:

```
1 %  
2 % Parameter of the Energy Management & Motion Control Strategy  
3 %  
4 %  
5 if (control_strategy == 1)  
6     flag.DYC = 0; flag.current = 0;  
7 end  
8 if (control_strategy == 2)  
9     flag.DYC = 1; flag.current = 1;  
10 end  
11 % "1": Baseline strategy (BS):  
12 % The vehicle is controlled by active front steering  
13 % The torque distribution ratios: tdis_FR = tdis_FL = 0.27;  
14 %
```

Remark 1:

- ❖ Modify “**meso_para.m**” so that the proposed strategy is selected and executed in the simulation when “**control_strategy == 2**”.

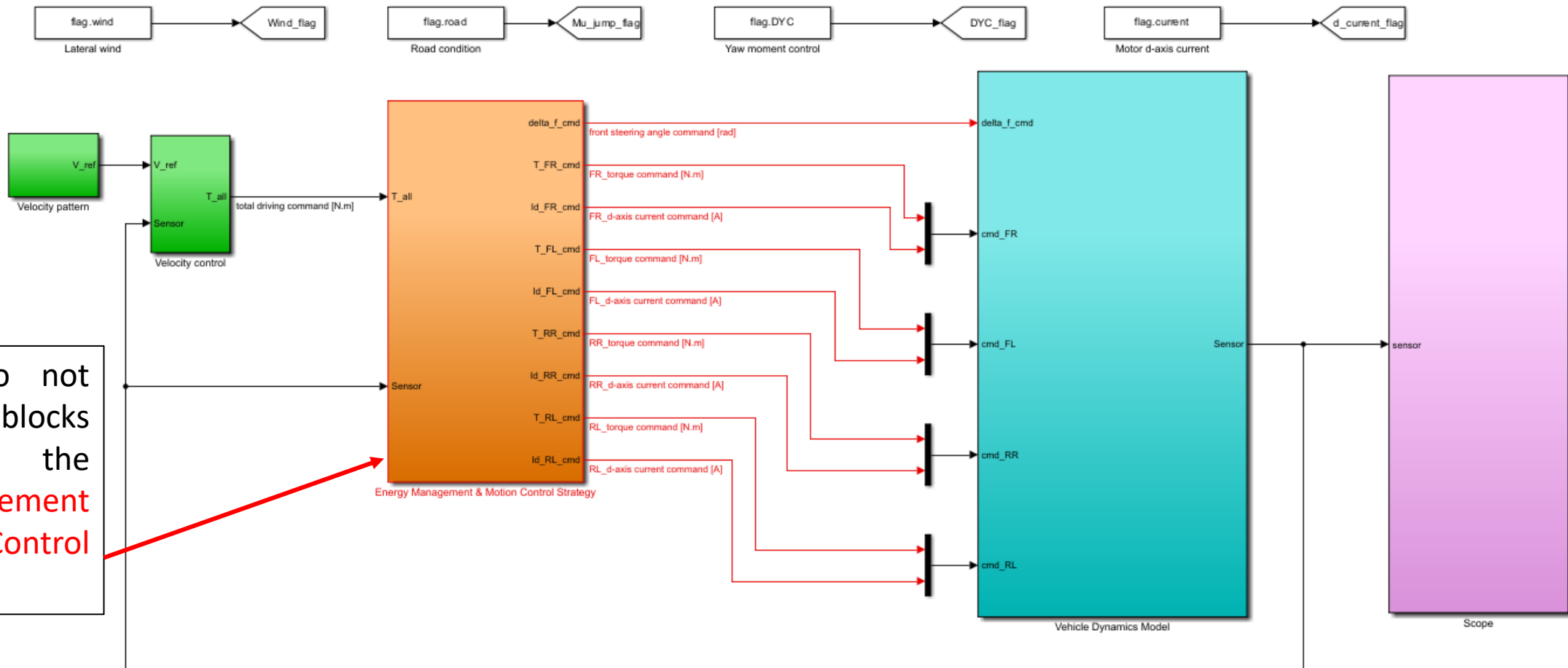
Competition Rules

```
27 %
28 %
29 %
30 % 1. Active front steering
31 Steer.Kp_pos = 0.02; % P gain of the lateral position control via front steering
32 Steer.Kp_yaw = 0.2; % P gain of the yaw angle control control via front steering
33 %
34 % 2. Yaw moment control (Improved strategy)
35 DYC.Kp_pos = 4000; % P gain of lateral position control via DYC
36 DYC.Ki_pos = 20; % I gain of lateral position control via DYC
37 DYC.Kd_pos = 1000; % D gain of lateral position control via DYC
38 DYC.Taud_pos = 0.005; % Time constant of Derivative terms
39 DYC.Kp_yaw = 5000; % P gain of yaw angle control via DYC
40 %
41 % 3. Torque distribution (Baseline strategy)
42 tdis_FR = 0.27; % torque distribution gain (front right)
43 tdis_FL = 0.27; % torque distribution gain (front left)
44 tdis_RR = 0.23; % torque distribution gain (rear right)
45 tdis_RL = 1 - (tdis_FR + tdis_FL + tdis_RR); % torque distribution gain (rear left)
46 %
```

Remark 2:

- ❖ Participants are free to modify “meso_para.m” as needed to implement their proposed strategies.
- ❖ The current parameters are provided only as examples. Participants are free to modify the control gains or develop alternative methods for **steering angle control, torque distribution, and d-axis current control**.
- ❖ The proposed strategy *must not* directly access information and parameters defined in LambdaMethod.m and driving condition.m. The strategy may use only the parameters explicitly permitted by the rules and the available sensor measurements to estimate/predict the unknown parameters.

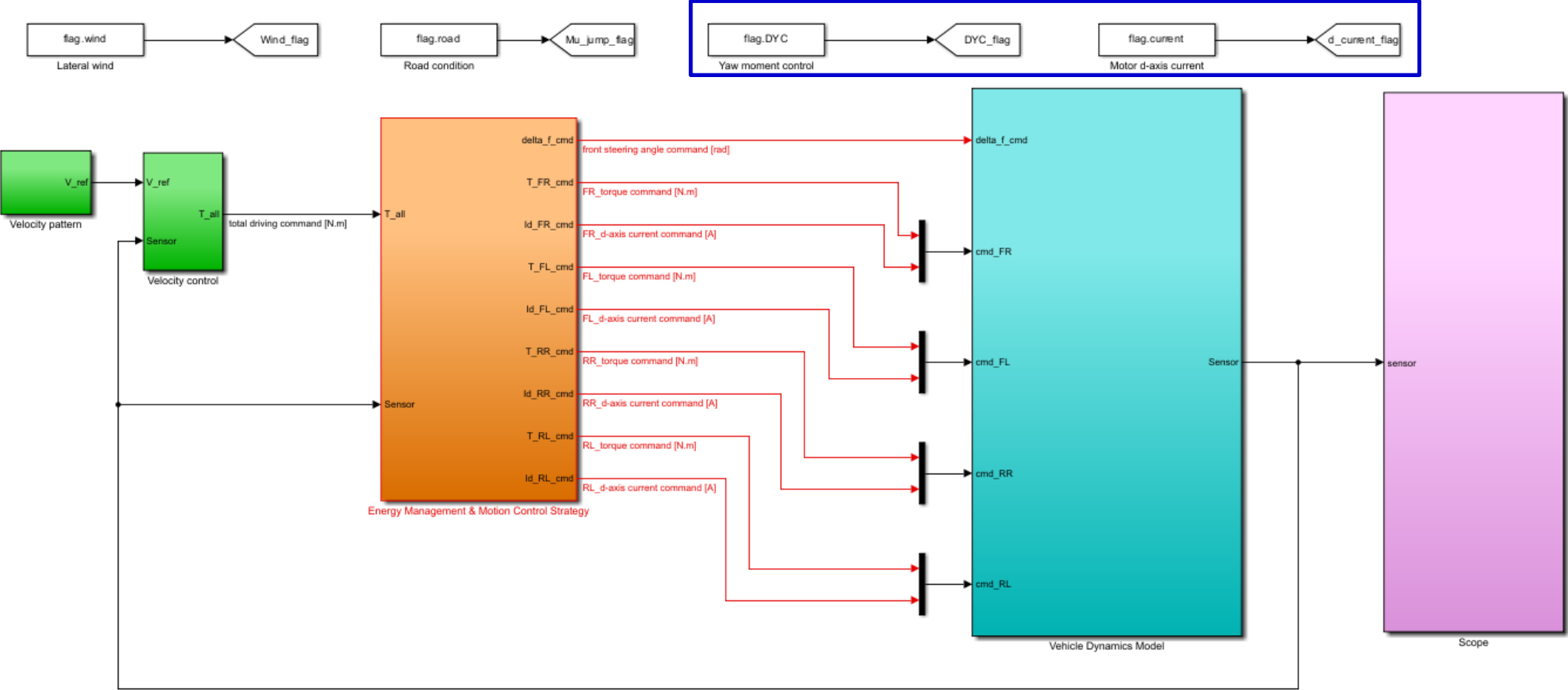
Competition Rules



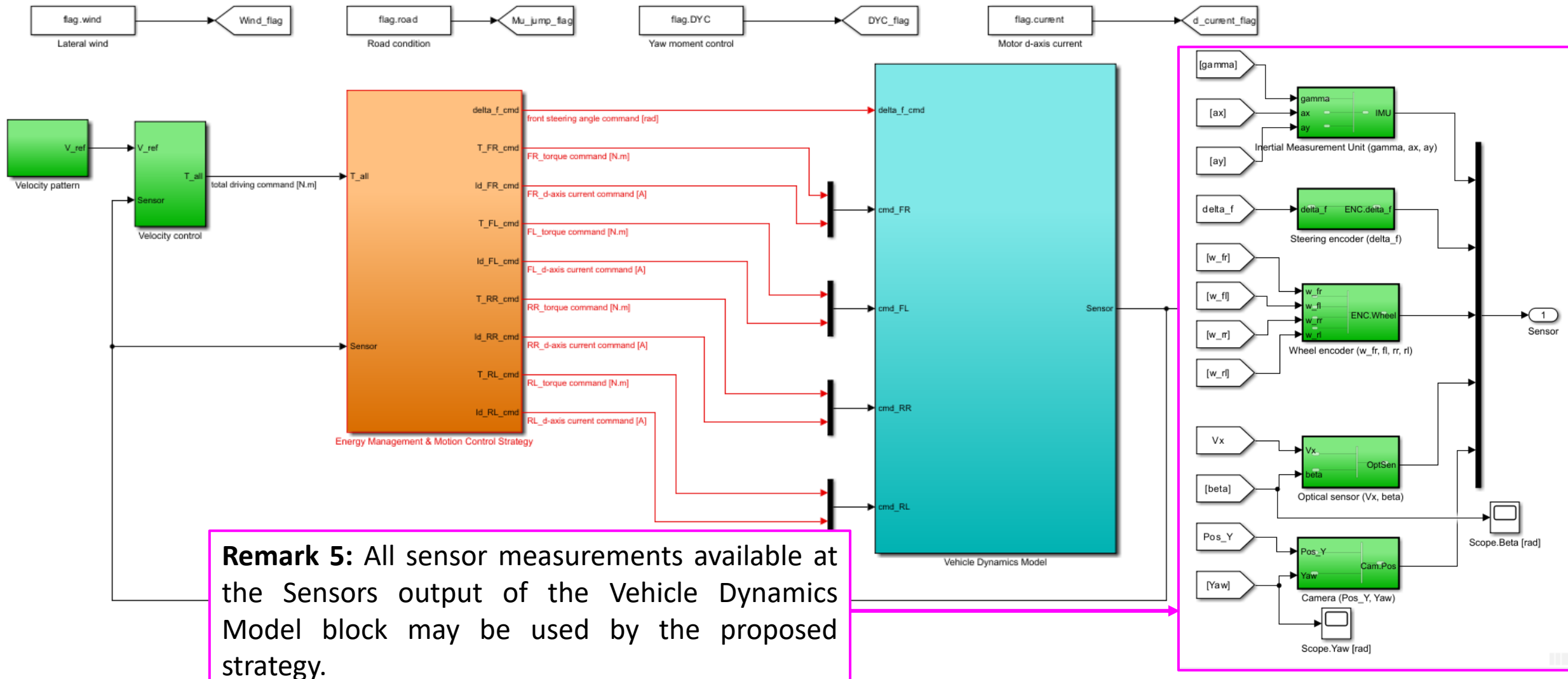
Remark 3: Do not modify any blocks other than the “Energy Management & Motion Control Strategy” block.

Competition Rules

Remark 4: The “DYC_flag” and “d_current_flag” may be used and modified as needed. Additional flags may also be defined in accordance with the proposed control strategy.



Competition Rules



Competition Rules

Remark 6: For each driving cycle, the score will be calculated as the sum of the scores obtained under the following driving conditions:

- ❖ **Condition 1** {flag.wind = 0; flag.road = 0}: high friction surface *without* lateral wind disturbance
- ❖ **Condition 2** {flag.wind = 1; flag.road = 0}: high friction surface *with* lateral wind disturbance
- ❖ **Condition 3** {flag.wind = 0; flag.road = 1}: friction variation surface *without* lateral wind disturbance
- ❖ **Condition 4** {flag.wind = 1; flag.road = 1}: friction variation surface *with* lateral wind disturbance

$$\phi_{\text{example}} = \sum_{j=1}^4 \phi_{\text{example}}(\text{jth condition})$$

$$\phi_{\text{secret}} = \sum_{j=1}^4 \phi_{\text{secret}}(\text{jth condition})$$



$$\phi_{\text{final}} = \frac{\phi_{\text{example}} + \phi_{\text{secret}}}{2}$$

Agenda

- ❑ Introduction
- ❑ Motivation
- ❑ Problem setting of MVC2027
- ❑ MATLAB/Simulink files
- ❑ Demonstration of MESO strategies
- ❑ GitHub repository
- ❑ **Registration and Submission**

Registration

IEEE VTS Motor Vehicle Challenge 2027 (MVC2027) - Resigtration Form

nguyen.binhminh@edu.k.u-tokyo.ac.jp [アカウントを切り替える](#)

* 必須の質問です

メール *

☒ 返信に表示するメールアドレスとして nguyen.binhminh@edu.k.u-tokyo.ac.jp を記録する

Team Name *

回答を入力

Name of Member 1 (contact person) *

回答を入力

Surname of Member 1 (contact person) *

Check this box
to receive the
confirmation
email.

Please enter the
team name,
contact email
address, and
names of all
team members,
and answer all
the questions.

Participants
must submit the
registration
form before
submitting the
proposed
strategies.

https://docs.google.com/forms/d/e/1FAIpQLScn9QM5dTbPJO58kUCpS6MpwM0aLW2bU7g9Atnztza7pp8_2uQ/viewform

What has to be uploaded for evaluation?

In order to evaluate each proposal, the teams must submit in the form the following files:

- ❖ “meso_para.m” renamed as “nameTeam_numProp_meso_para.m” where “numProp” must identify the proposal number of the team. If only one proposal is presented, please use “num1” (e.g., “teamABC_prop1_meso_para.m”). Please avoid spaces in the file name.
- ❖ “MVC27_Sim.slx” renamed as “nameTeam_numProp_MVC27_Sim.slx” where “numProp” must identify the proposal number of the team. If only one proposal is presented, please use “num1” (e.g., “teamABC_prop1_MVC27_sim.slx”). Please avoid spaces in the file name.
- ❖ Complete simulation package of the proposed solution (*backup*): Create a ZIP file including the following files: main.m, sim_para.m, nameTeam_numProp_meso_para.m, driving_condition.m, LambdaMethod.m, sim_result_plot.m, scoring.m, nameTeam_numProp_MVC27_Sim.slx.
- ❖ The score text file may also be included. Participants can also include a file that briefly explains their idea, and any instruction for the organizers to reproduce and evaluate the proposed strategy.
- ❖ Rename the ZIP file as “nameTeam_numProp_MVC27.zip”. Please note the ZIP file is only a backup.
Evaluation will mainly be based on the individual submitted files.

Submission

Check this box to receive the confirmation email.

IEEE VTS Motor Vehicle Challenge 2027 (MVC2027) - Submission Form

nguyen.binhminh@edu.k.u-tokyo.ac.jp
[アカウントを切り替える](#)  下書きを保存しました

ファイルをアップロードしてこのフォームを送信すると、Google アカウントに関連付けられている名前、メールアドレス、および写真が記録されます

* 必須の質問です

メール *

☒ 返信に表示するメールアドレスとして nguyen.binhminh@edu.k.u-tokyo.ac.jp を記録する

Team Name *

回答を入力

Name of Contact Person *

回答を入力

Surname of Contact Person *

回答を入力

Email of Contact Person *

回答を入力

Simulation files *

最大 5 個のサポートされているファイルをアップロードします。1 ファイルは最大 10 MB です。

 ファイルを追加

回答のコピーが nguyen.binhminh@edu.k.u-tokyo.ac.jp にメールで送信されます。

[フォームをクリア](#)

Please enter the team name, name of the contact person, and contact email address.

Upload files from your computer

- nameTeam_numProp_meso_para.m
- nameTeam_numProp_MVC27_Sim.slx
- nameTeam_numProp_MVC27.zip

Click this button to submit files

<https://docs.google.com/forms/d/e/1FAIpQLSdaAZOpIFS4j9ZSh1LC5OPYZo41tUQKE3v68uK8hWsC88eRkA/viewform>

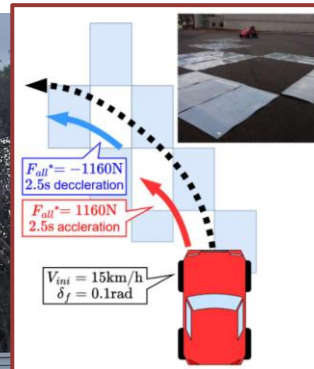
Secret driving cycle

- ❖ The secret driving cycle is established based on a **real-world driving pattern** recorded for a vehicle traveling from the University of Tokyo to Kashiwanoha-campus Station.
- ❖ The road friction varies randomly and non-uniformly between the left and right wheels.
- ❖ The vehicle may encounter low-friction surfaces and lateral wind disturbances simultaneously, including during acceleration and deceleration.

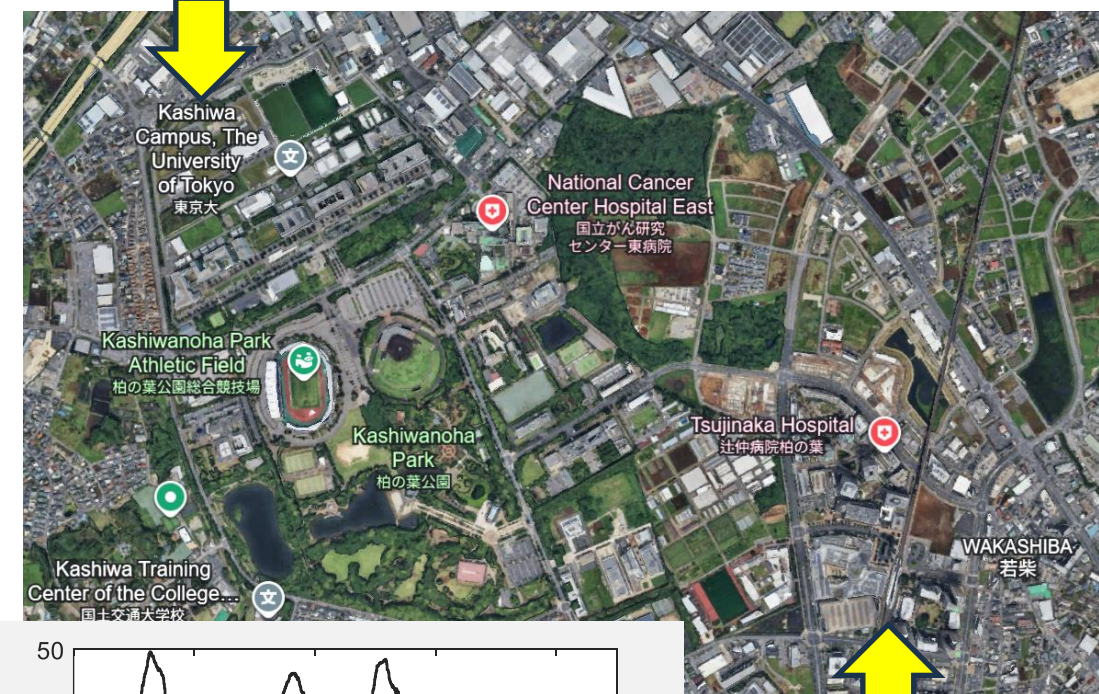


Example of road surface with random friction

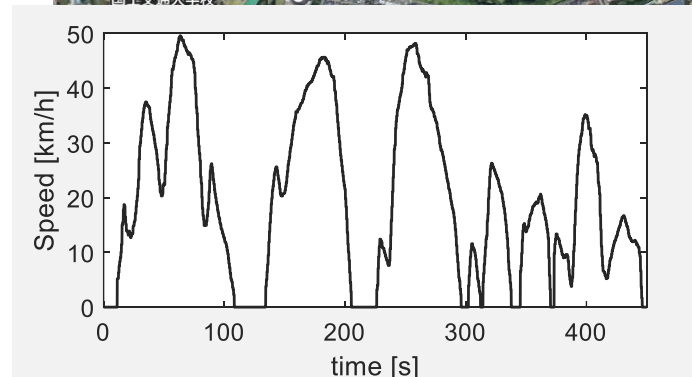
[\(Youtube link\)](#)



The University of Tokyo



Kashiwanoha Campus Station



Questions and Support

- ❑ Please post technical questions on GitHub.
- ❑ Please do not hesitate to contact us by email if you have any questions.

Binh-Minh Nguyen: nguyen.binhminh@edu.k.u-tokyo.ac.jp

Facundo Aguilera: faguilera@ing.unrc.edu.ar

- ❑ We might organize online meeting to discuss with you and provide technical support.

ENJOY THE CHALLENGE AND MAY THE BEST WIN

https://github.com/VTSociety/MVC_2027/



- Registration deadline: 31st January 2027
- Submission: 1st March 2027
- Results: 31 March 2027

**IEEE****VTS**
Connecting the Mobile World

IEEE Vehicular Technology Society

IEEE VTS Motor Vehicle Challenge 2027

**Management of Energy Efficiency and Safe Operation
for Multi-Actuated Ground Vehicles on DWPT Road**

The Motor Vehicle Challenge (MVC), supported by IEEE Vehicular Technology Society, is an annual activity to find an appropriate energy management strategy to improve electric vehicles' performance. In this eleventh edition of the event, the challenge focuses on the energy-efficient and safe operation of an in-wheel-motor vehicle with dynamic wireless power transfer operating on a road subject to various disturbances. The competition is open to everyone, students, academics, and industry.

DWPT CHARGING COILS

CHALLENGE

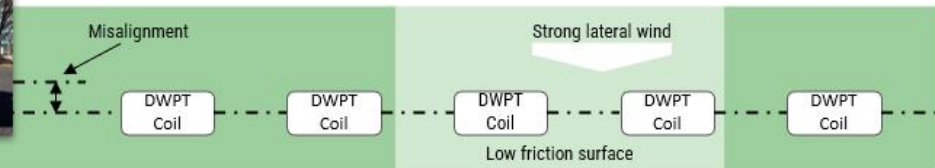
This challenge considers a four in-wheel-motor vehicle (4IWM-EV) operating autonomously on roads with a dynamic wireless power transfer (DWPT) infrastructure, based on prototypes developed at the University of Tokyo. Under realistic conditions, the vehicle is subjected to abrupt variations in road friction due to weather changes, as well as strong lateral wind disturbance. Participants are therefore invited to develop an integrated strategy that coordinates motor torque distribution, d-axis current control, and front steering angle.

The MVC2027 intends to serve as a platform for discussing and advancing the concept of MESO (Management of Energy Efficiency and Safe Operation) for multi-actuated vehicles.

SCORING

Participants of the Challenge will be provided with a model of the 4IWM-EV and DWPT platform on MATLAB/Simulink, together with sample control strategies and speed profiles. Each proposed strategy will be evaluated using a scoring function that assesses the achievement of multiple objectives, including:

- improving overall drivetrain efficiency;
- ensuring safe traction by regulating wheel slip ratios;
- enhancing infrastructure utilization by maximizing the energy received from DWPT;
- improving driving comfort by minimizing deviations in lateral acceleration.



Misalignment

Strong lateral wind

Low friction surface

4IWM-EV

DWPT Coil

DWPT Coil

DWPT Coil

DWPT Coil

DWPT Coil